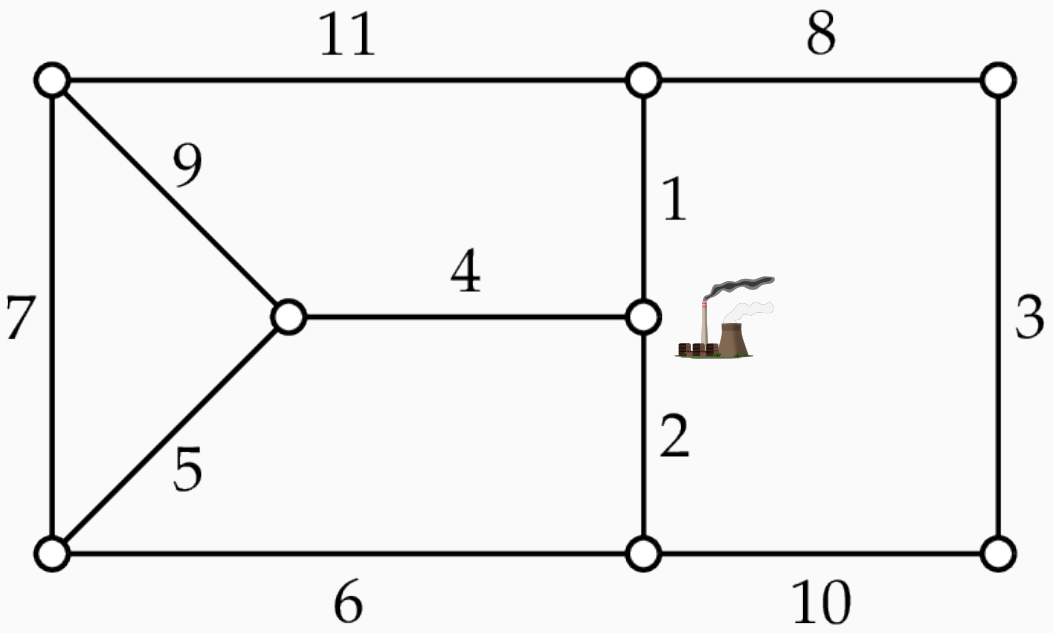
Minimum Spanning Trees

Prof. Timothy Sun

Setting up a power grid

Goal: add power lines to give power to each city



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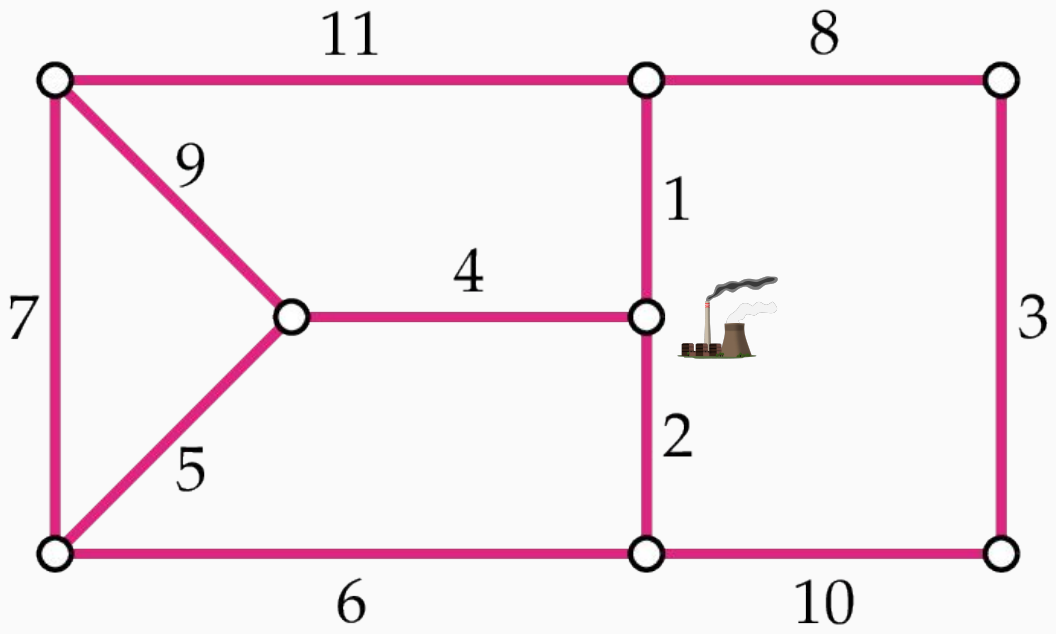
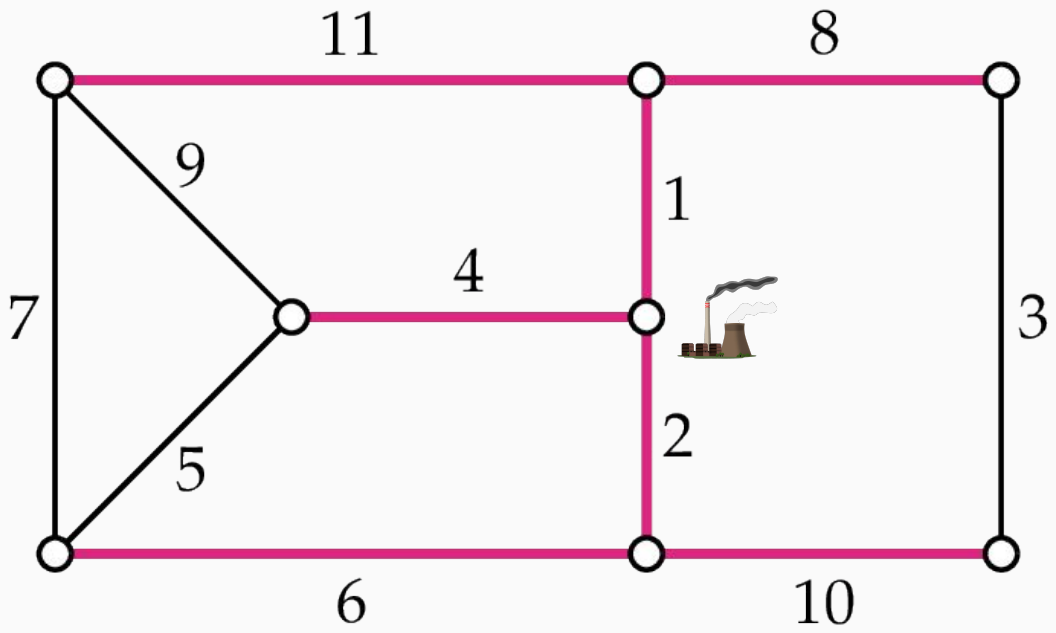


Setting up a power grid

Goal: add power lines to give power to each city, ***minimizing cost***



Only need to include some of the edges to ***maintain connectivity***.

Setting up a power grid

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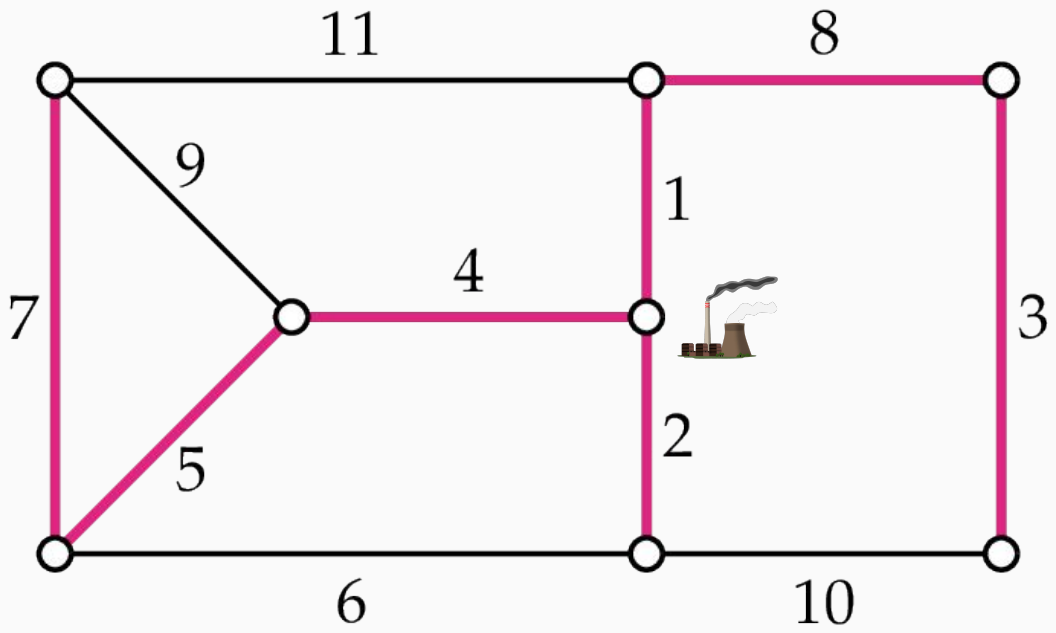

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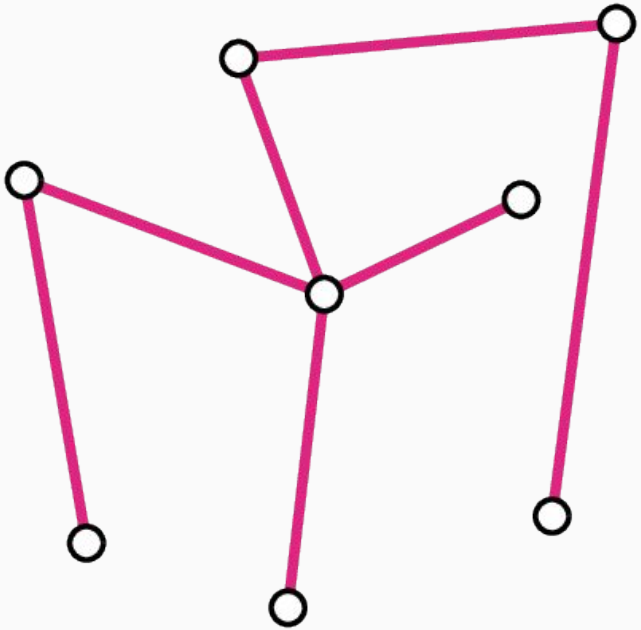
Trees

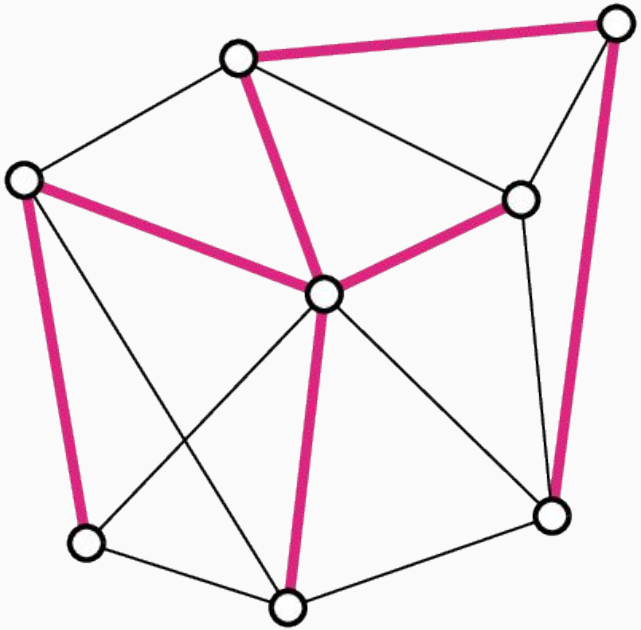
A **tree** is an undirected, ***acyclic***, ***connected*** graph.



Trees

A **tree** is an undirected, ***acyclic***, ***connected*** graph.

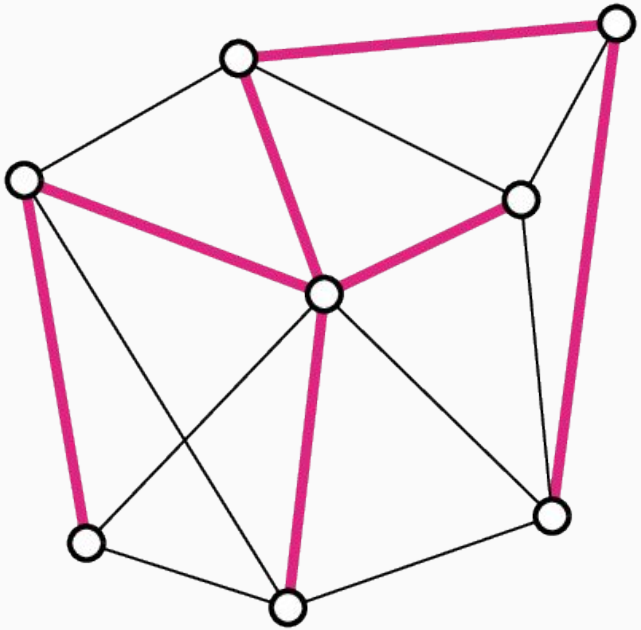
A **spanning tree** is a subgraph on the same vertex set that's a tree.



Trees

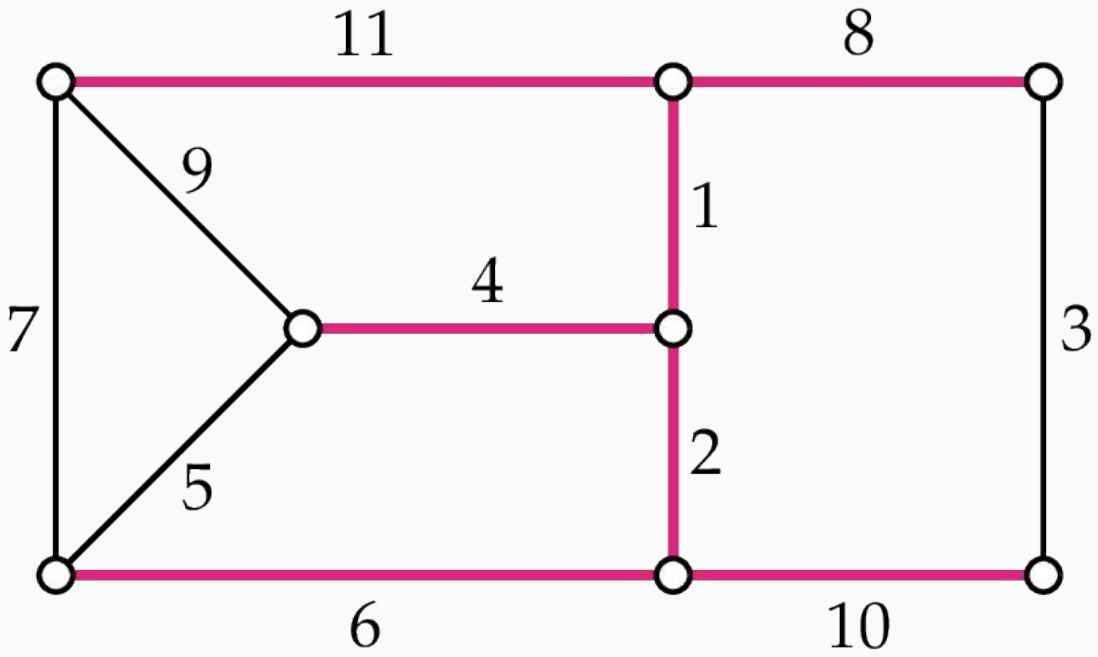
Properties:

- “Minimal connected graph”
- “Maximal acyclic graph”
- $|\boldsymbol{E}| = |\boldsymbol{V}|-1$
- Unique path between pairs of vertices
- etc.



Minimum Spanning Tree

Each edge has a **weight** $w(\mathbf{e})$



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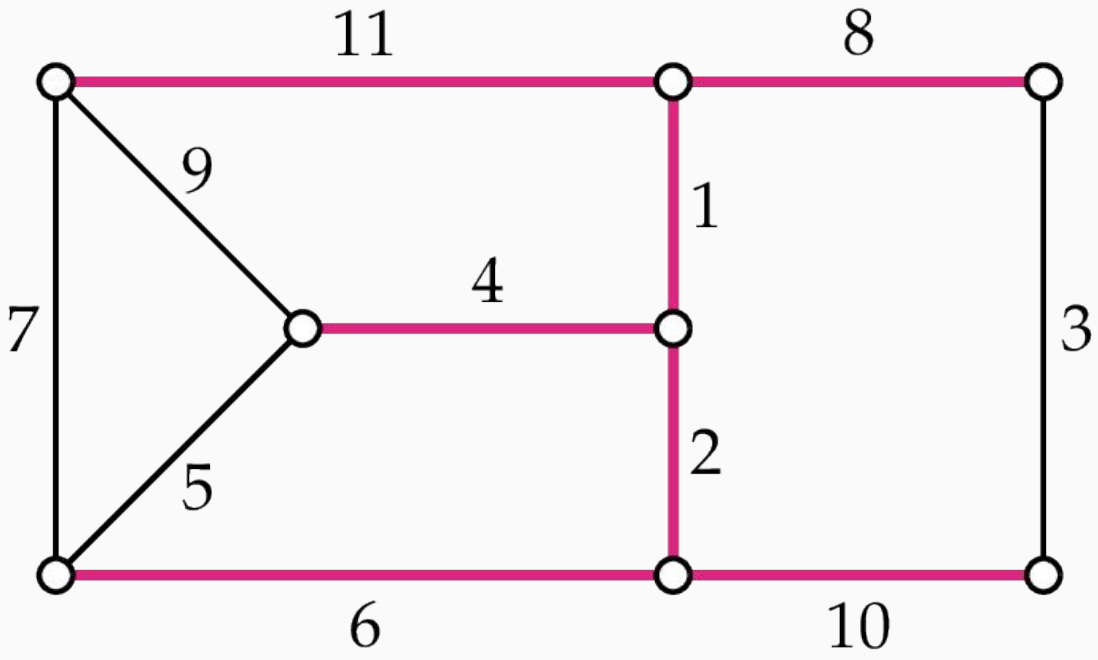
The weight of a tree $w(\boldsymbol{T})$ is the sum of its edge weights



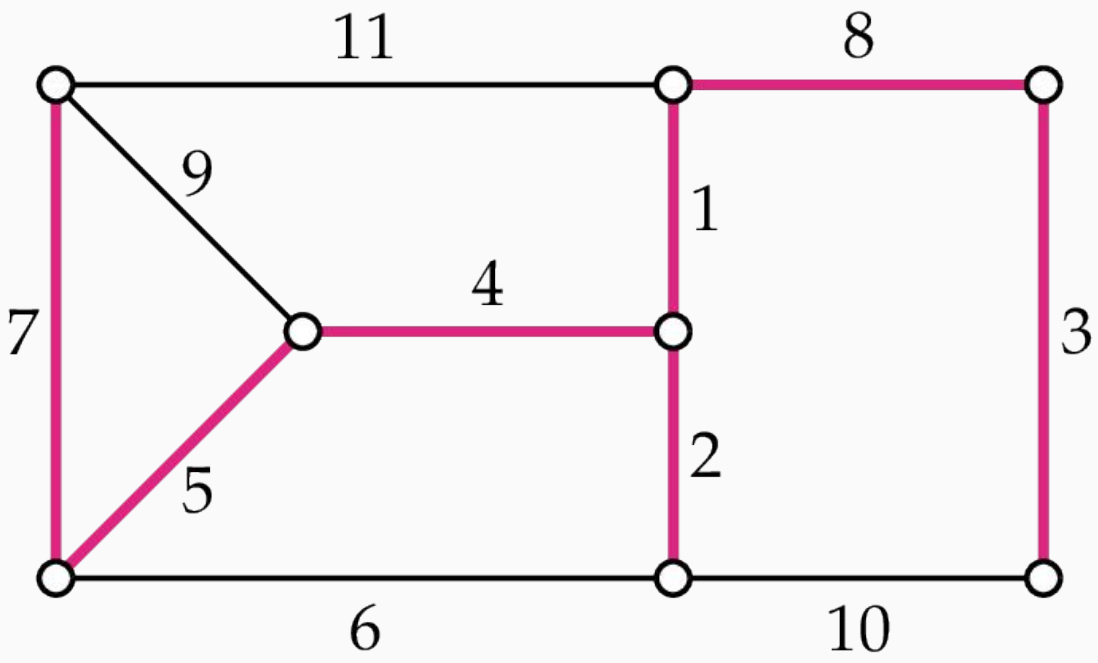
$w(\boldsymbol{T}) = 42$

Minimum Spanning Tree

Each edge has a **weight** $w(\boldsymbol{e})$

The weight of a tree $w(\boldsymbol{T})$ is the sum of its edge weights

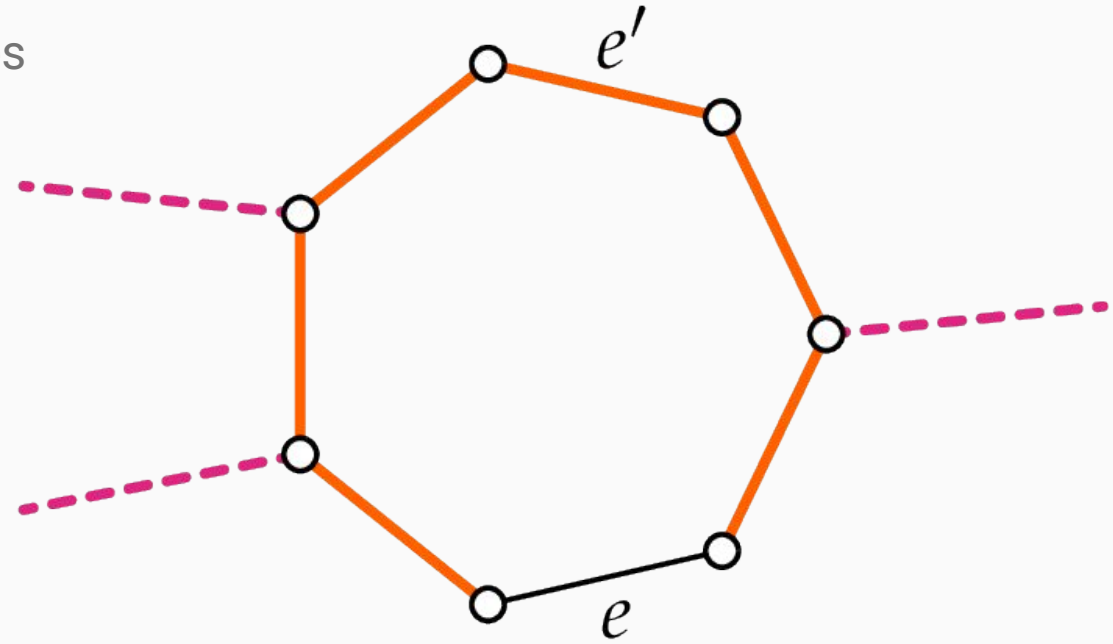
A ***minimum spanning tree*** is a spanning tree with smallest weight



$w(\boldsymbol{T}) = 30$

Fundamental cycles

Adding any edge to a tree makes a **fundamental cycle**.



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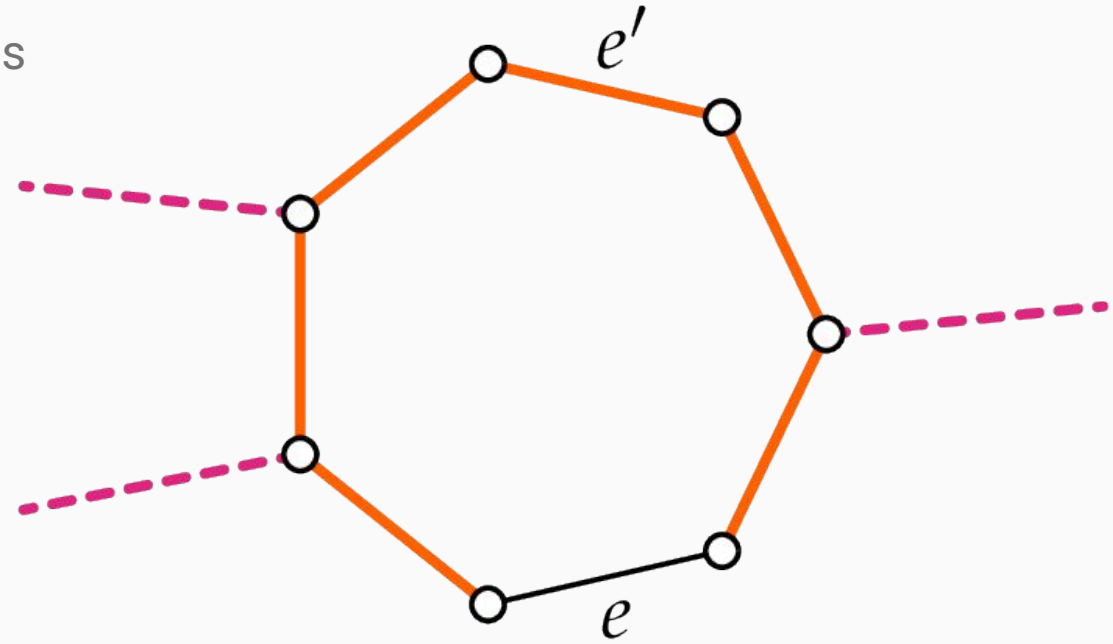
What can we say about $w(\mathbf{e})$ if the tree is an MST?



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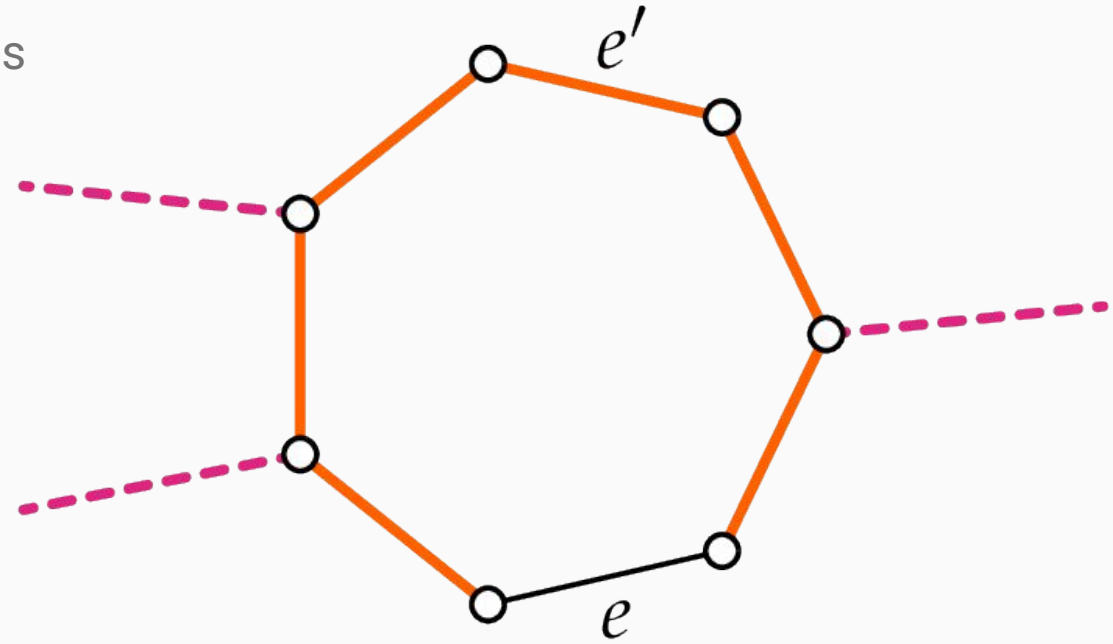
For every e' on fundamental cycle, $w(e') \leq w(e)$



Fundamental cycles

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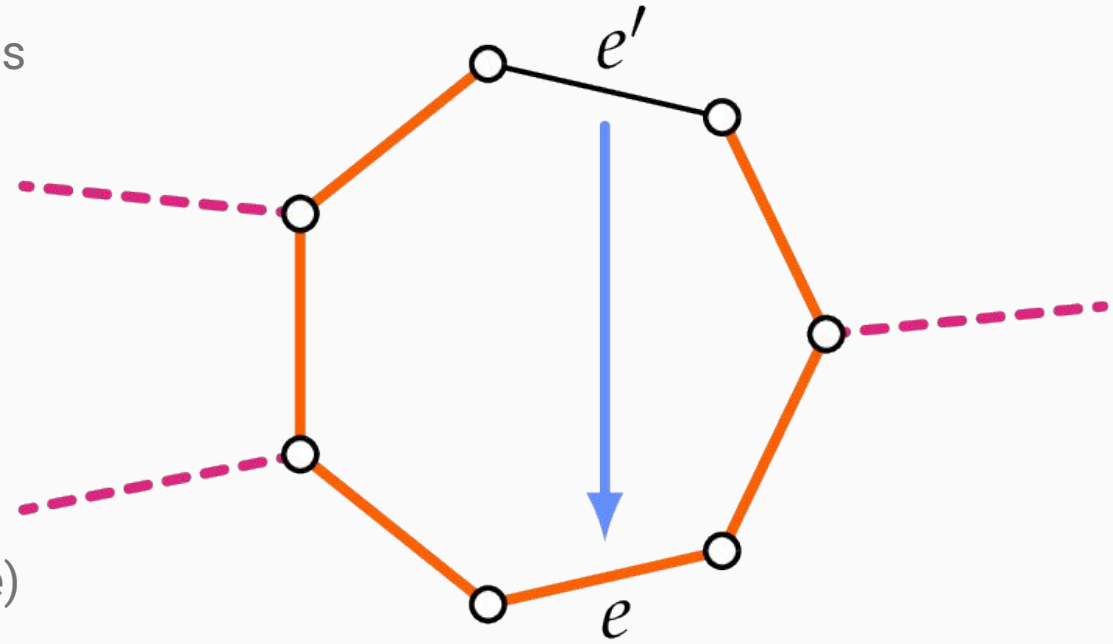
What can we say about w(**e**) if the tree is an MST?

For every e' on fundamental cycle, *w*(e') ≤ *w*(e)
(otherwise, you could **exchange**)



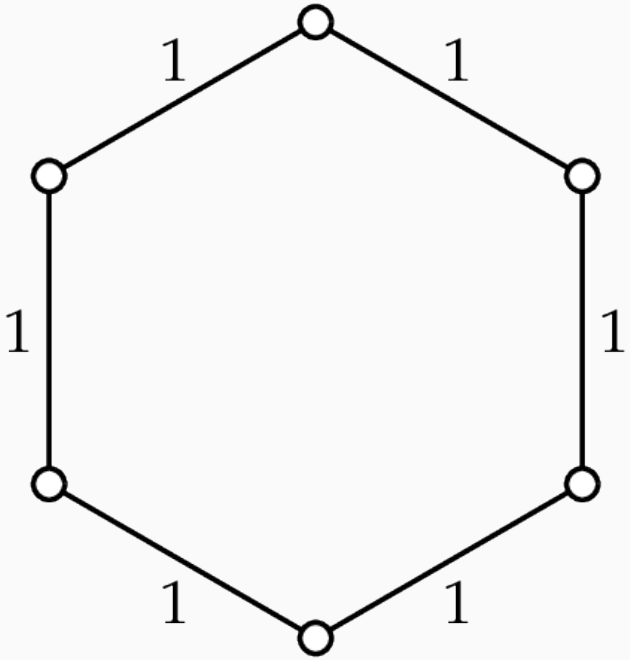
MST Uniqueness

Useful to think of ***the*** MST of a graph…

MST Uniqueness

Useful to think of ***the*** MST of a graph…

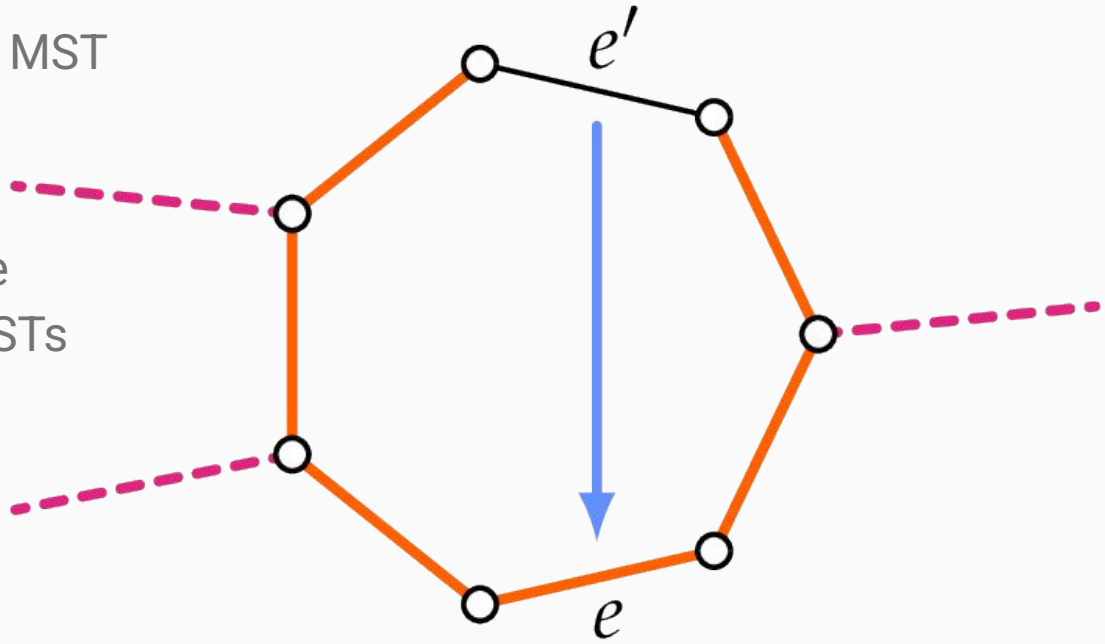
...but there can be more than one.



MST Uniqueness

Distinct edge weights → unique MST

Proof sketch: fundamental cycle flipping between two distinct MSTs



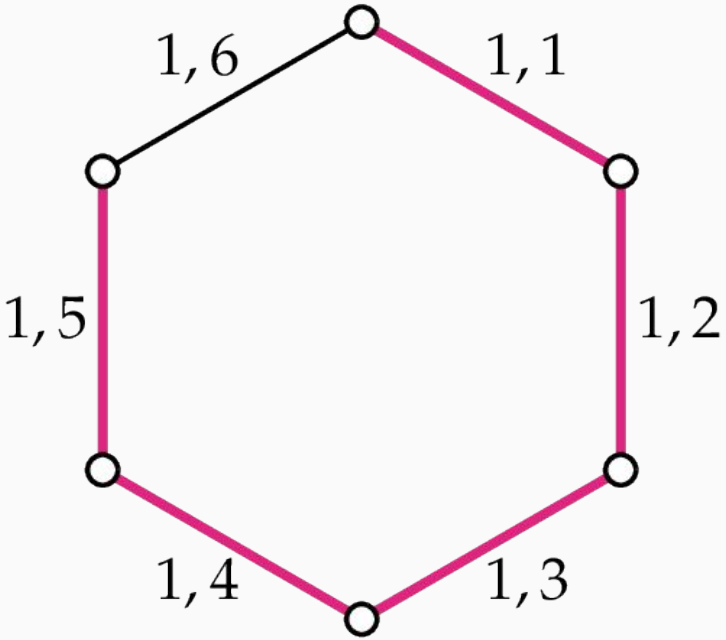
MST Uniqueness

What about duplicate edge weights?

Find a consistent tie-breaking rule, e.g.:

- Order the edges
- Perturb the edge weights

MST Uniqueness



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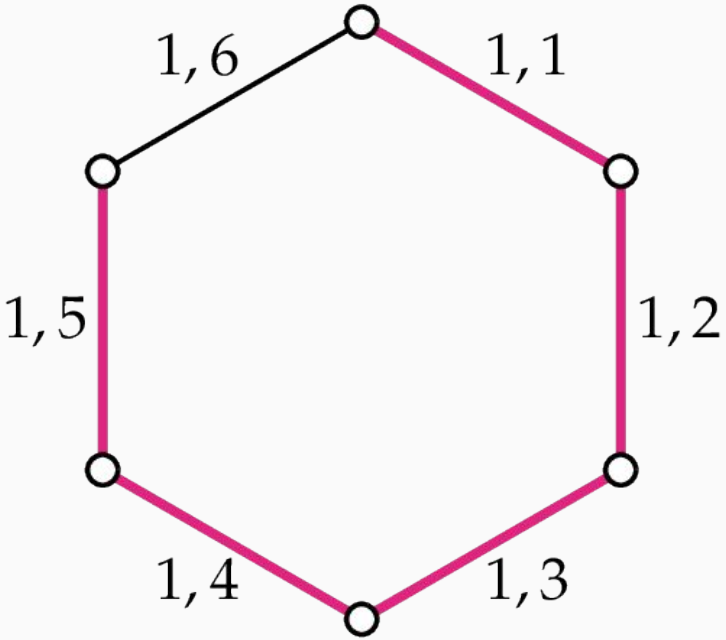
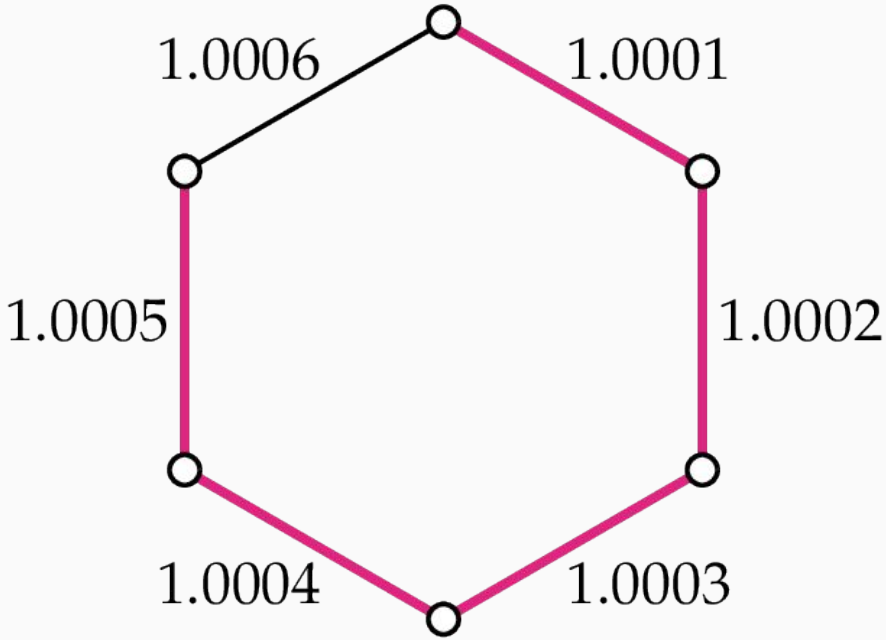
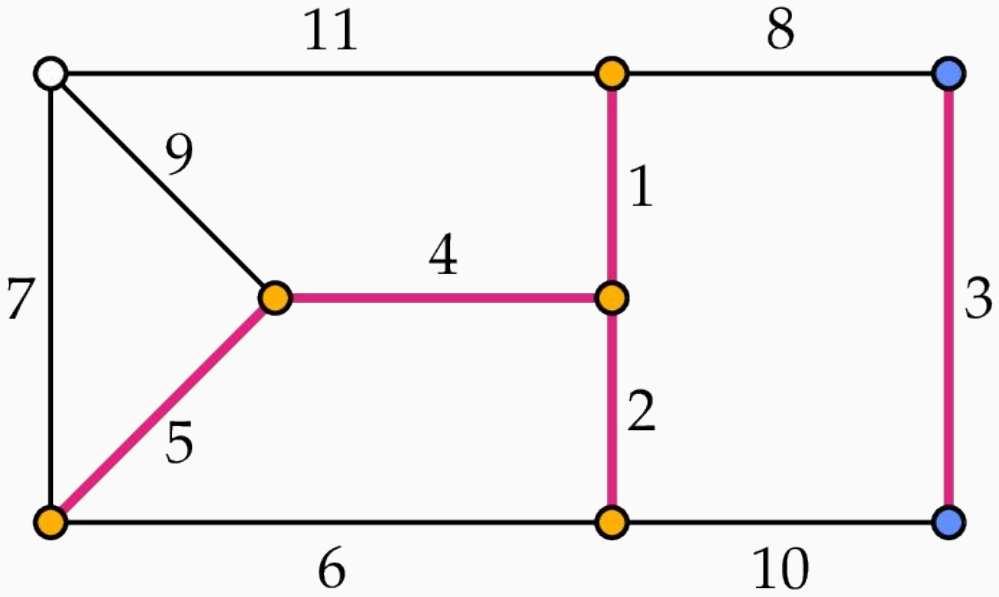




Safe Edges

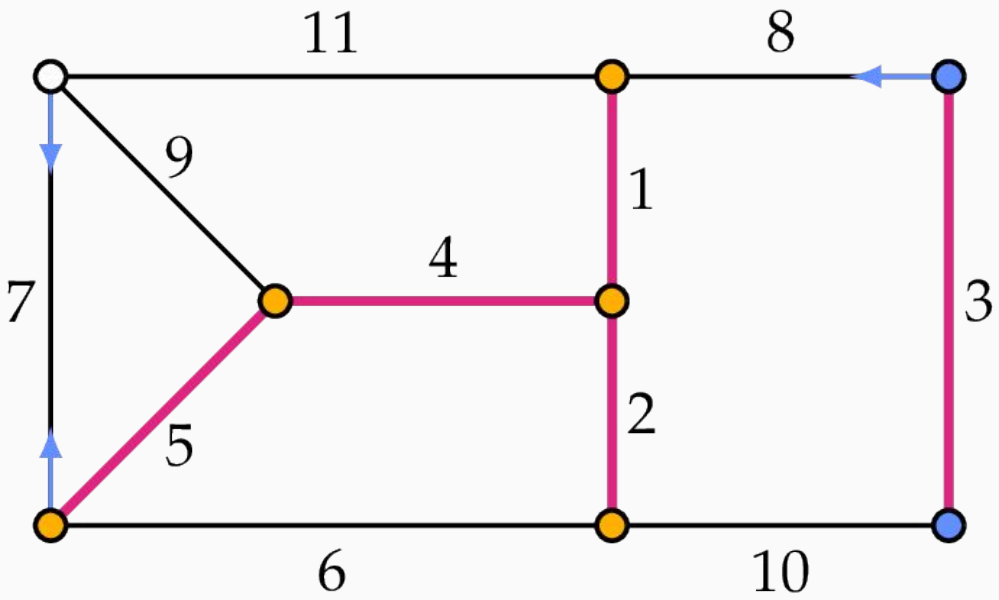
At every iteration, an MST algorithm maintains a “spanning forest” ***F***



Safe Edges

At every iteration, an MST algorithm maintains a “spanning forest” ***F***

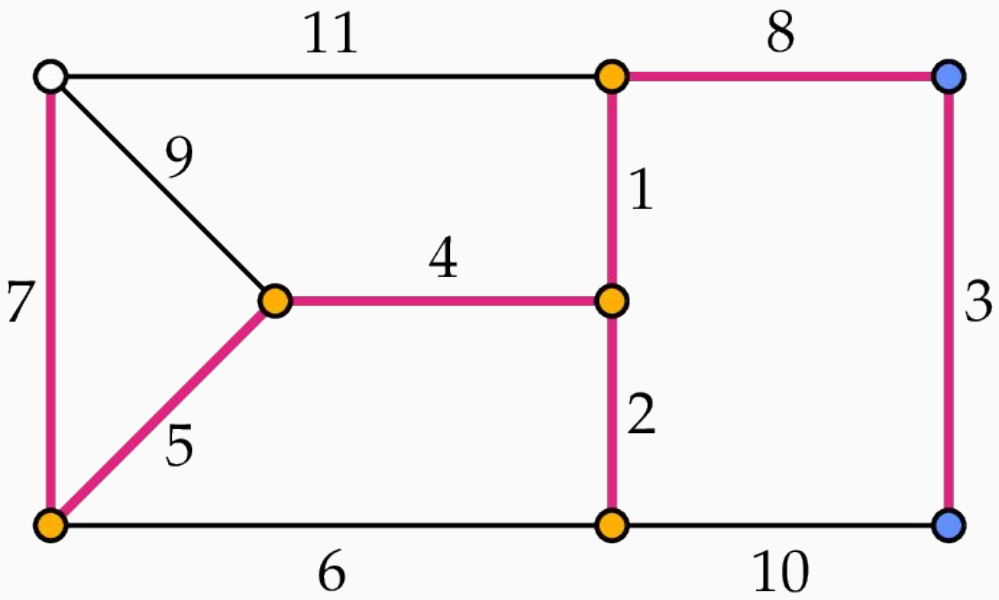
For a component of ***F***, its **safe edge** is the ***lightest*** edge going to a different component



Safe Edges

General idea: “find a safe edge and add to ***F***” (greedy)

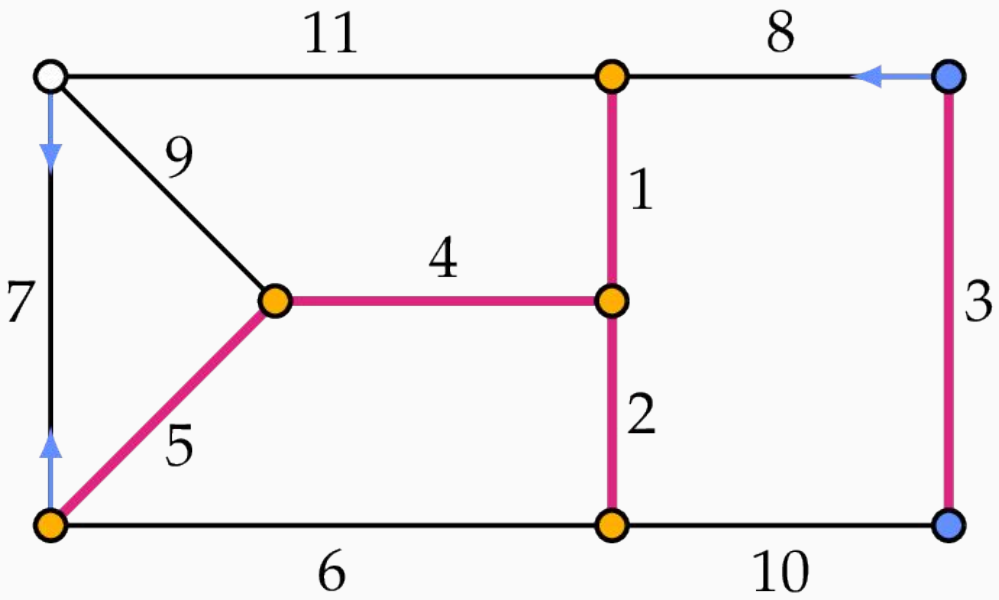
Claim: MST contains all safe edges



Safe Edges

Executive summary:

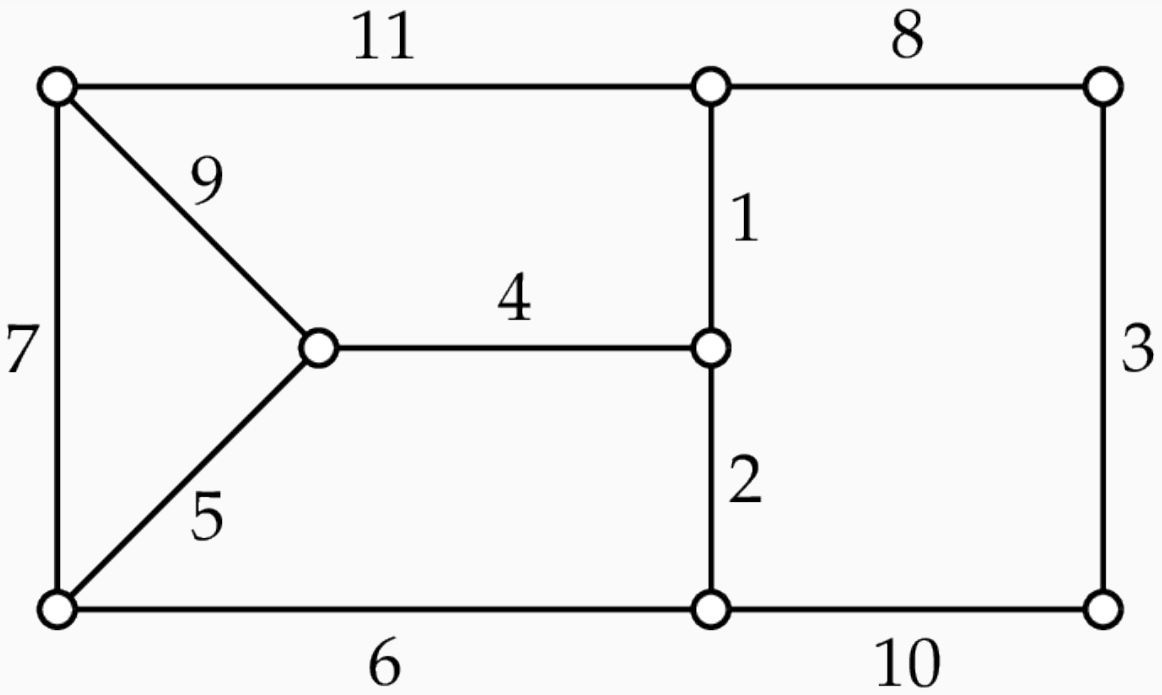
- Kruskal: add lightest safe edge
- Prim: add safe edge of component of ***v***
- Boruvka: add all safe edges at once



Kruskal’s algorithm

1. Sort all the edge weights.
2. Repeatedly pick the smallest edge without creating a cycle.

Kruskal’s algorithm



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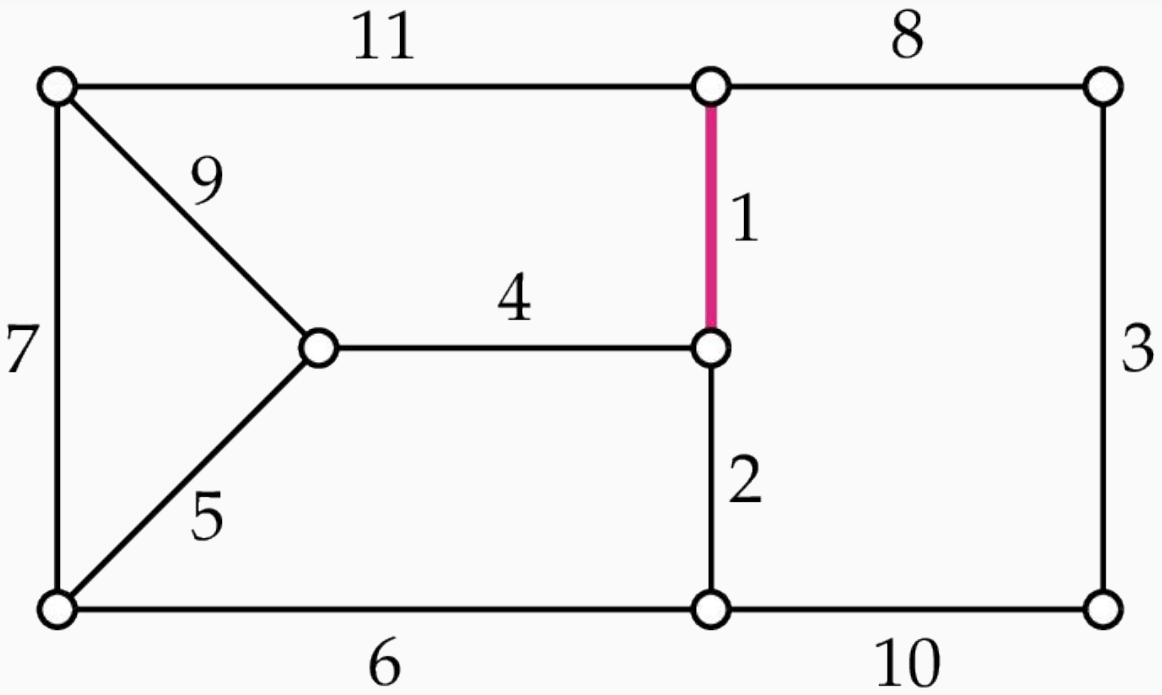

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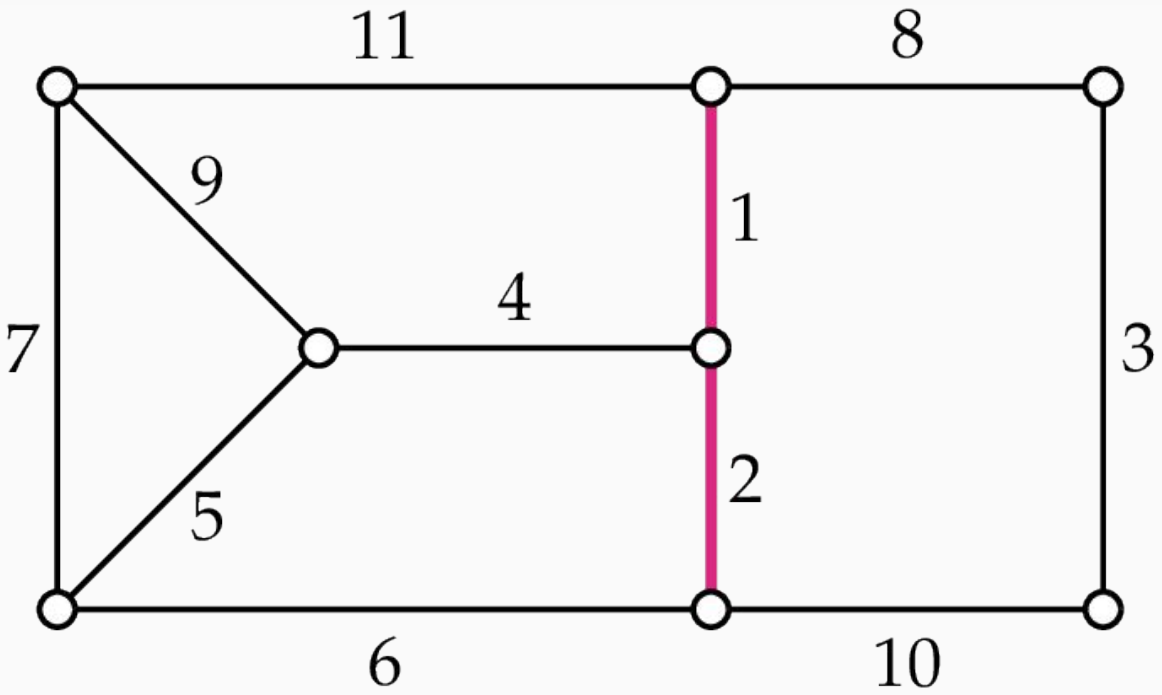

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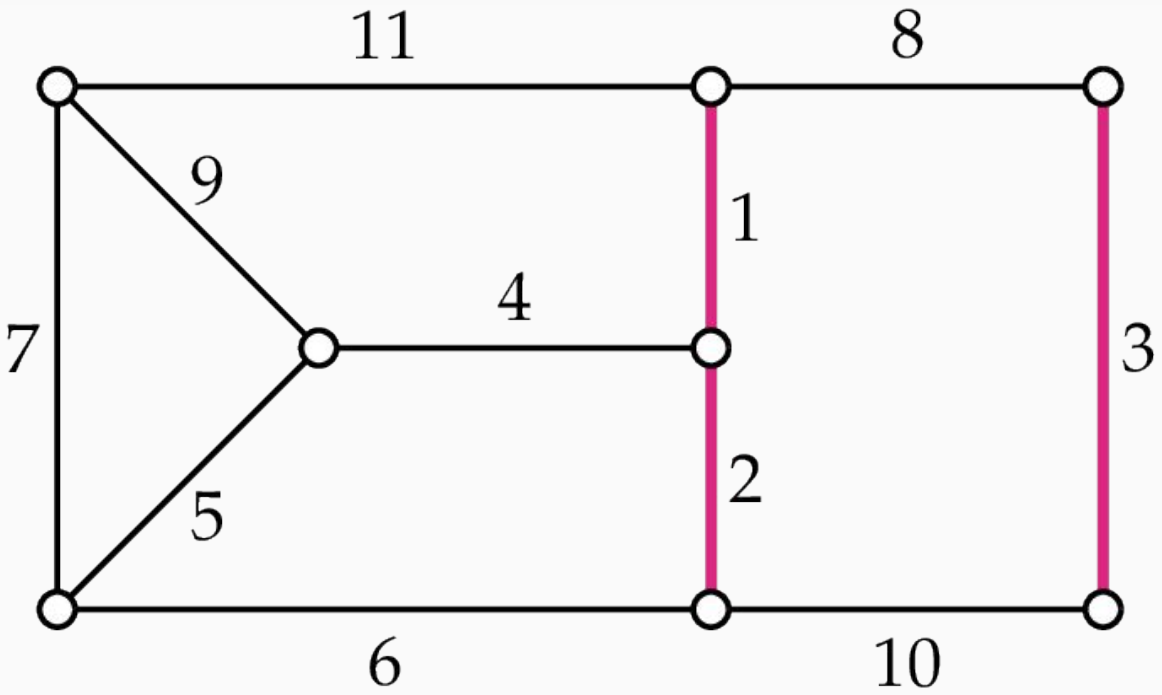

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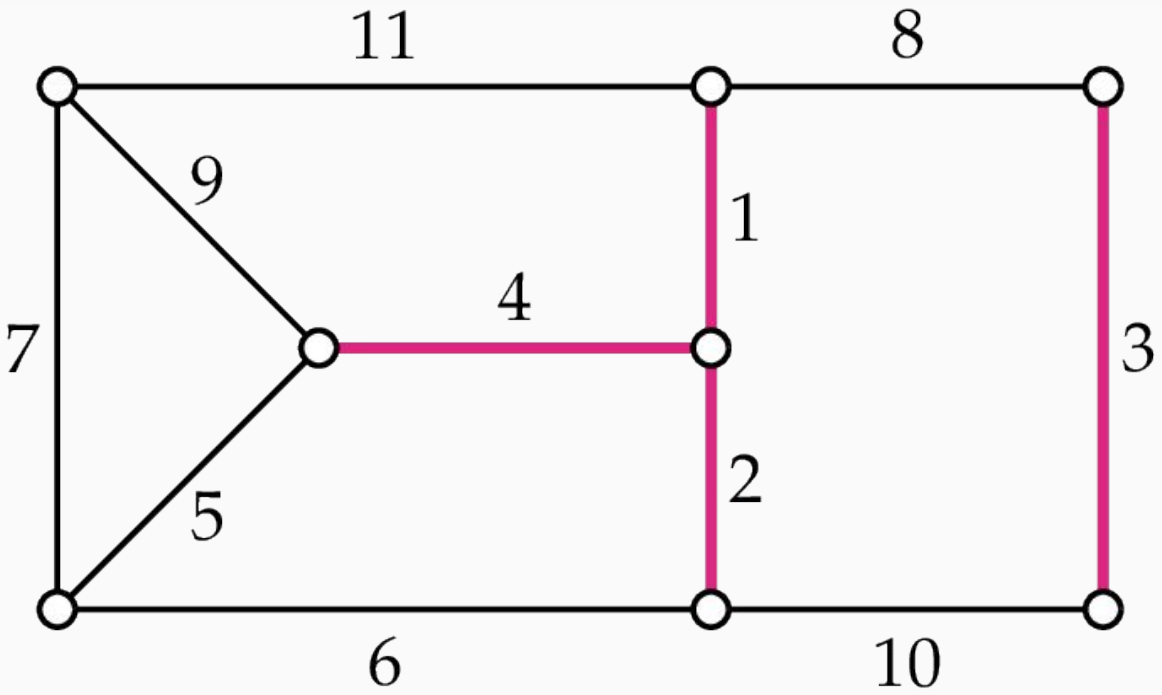

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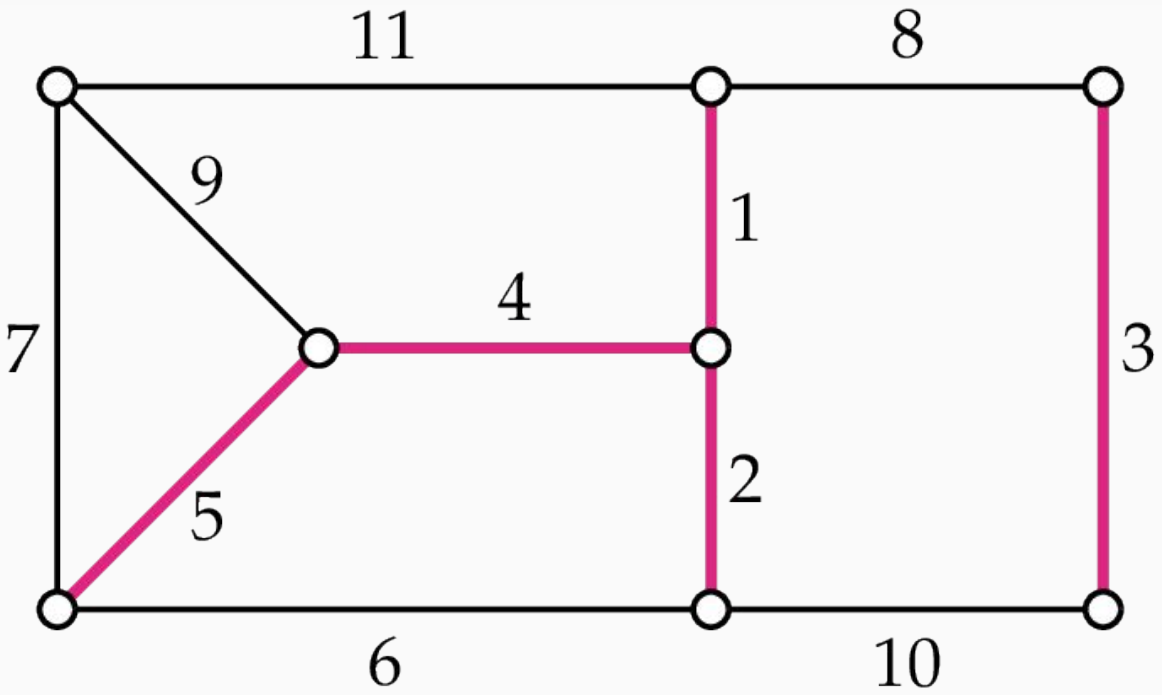

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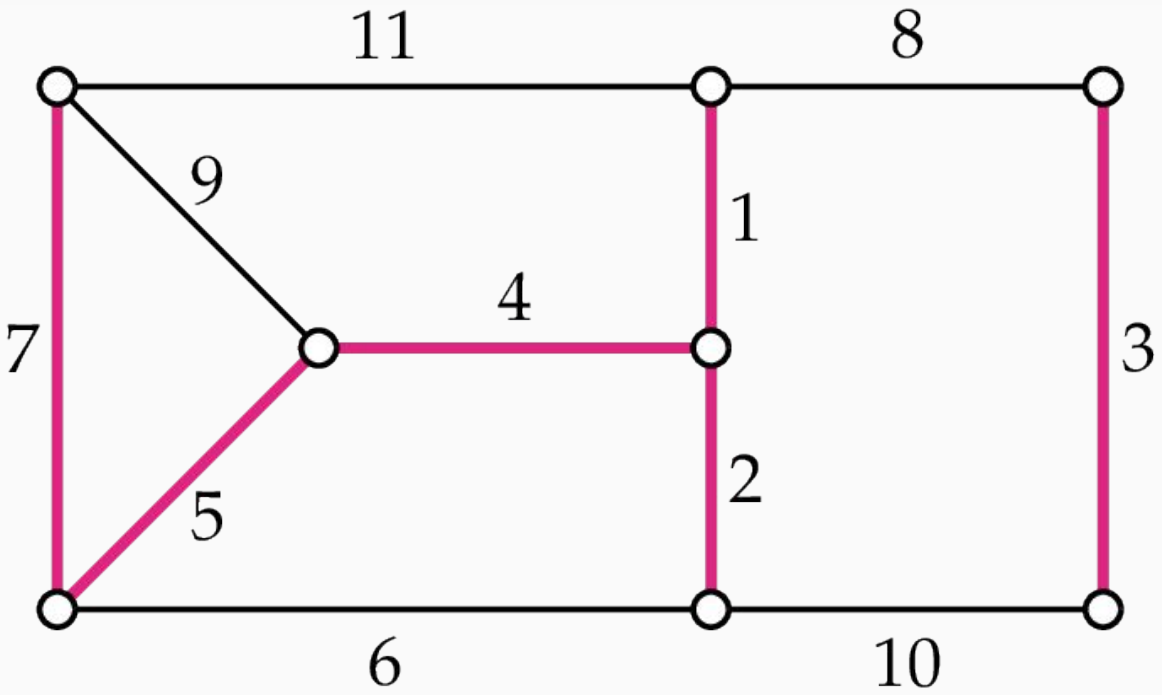

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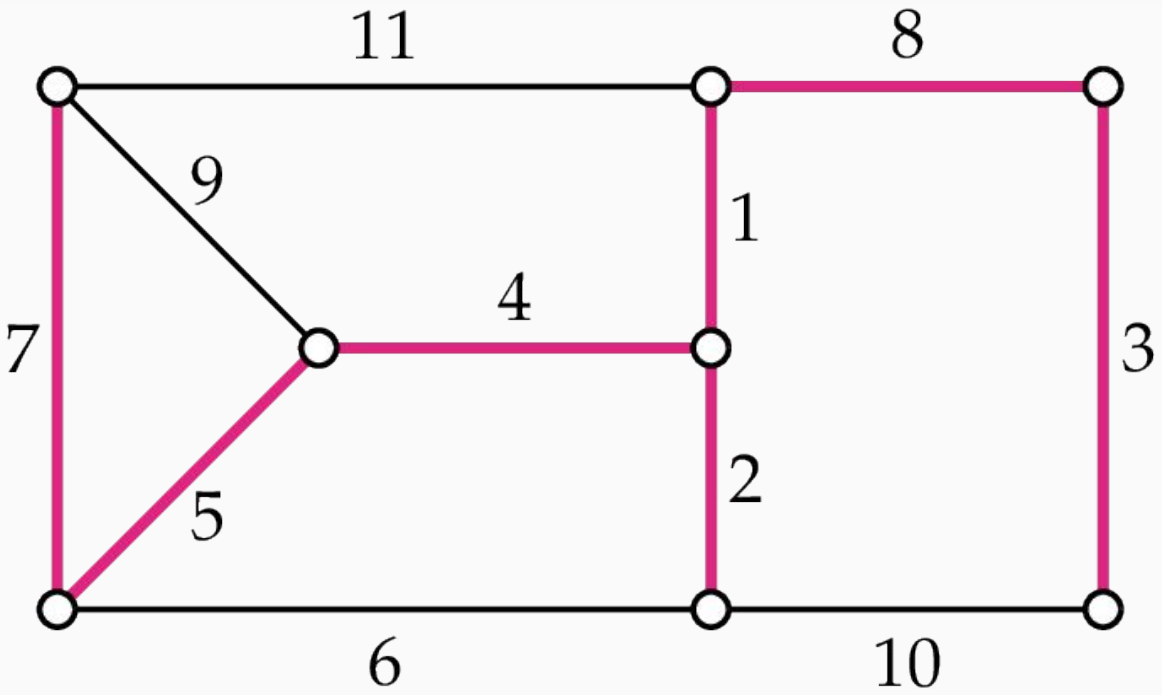
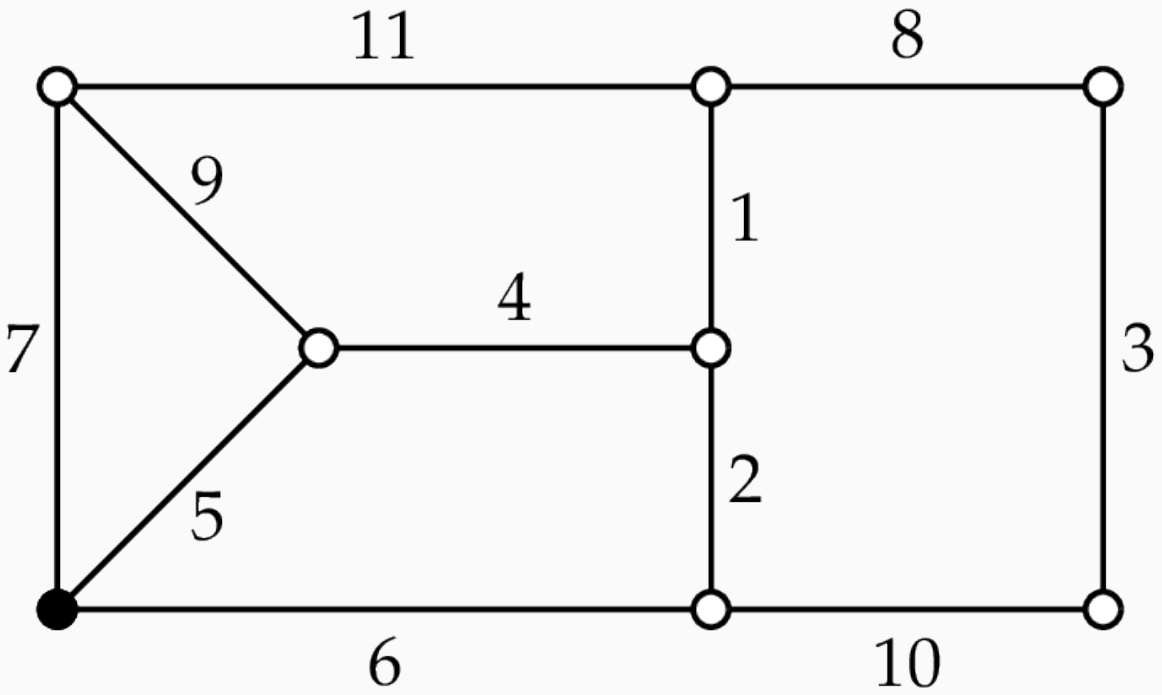


Prim’s algorithm

1. Pick any vertex **v**, and set **T** = {**v**}
2. Repeatedly “grow” **T** by adding smallest edge leaving **T**

Prim’s algorithm



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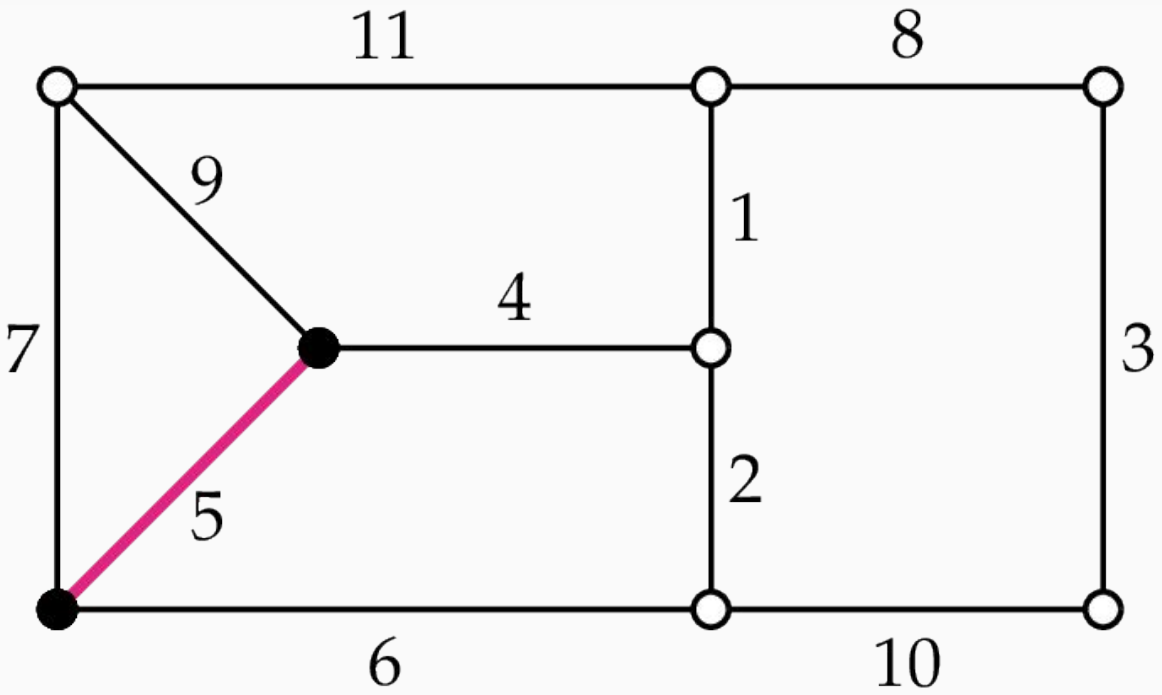

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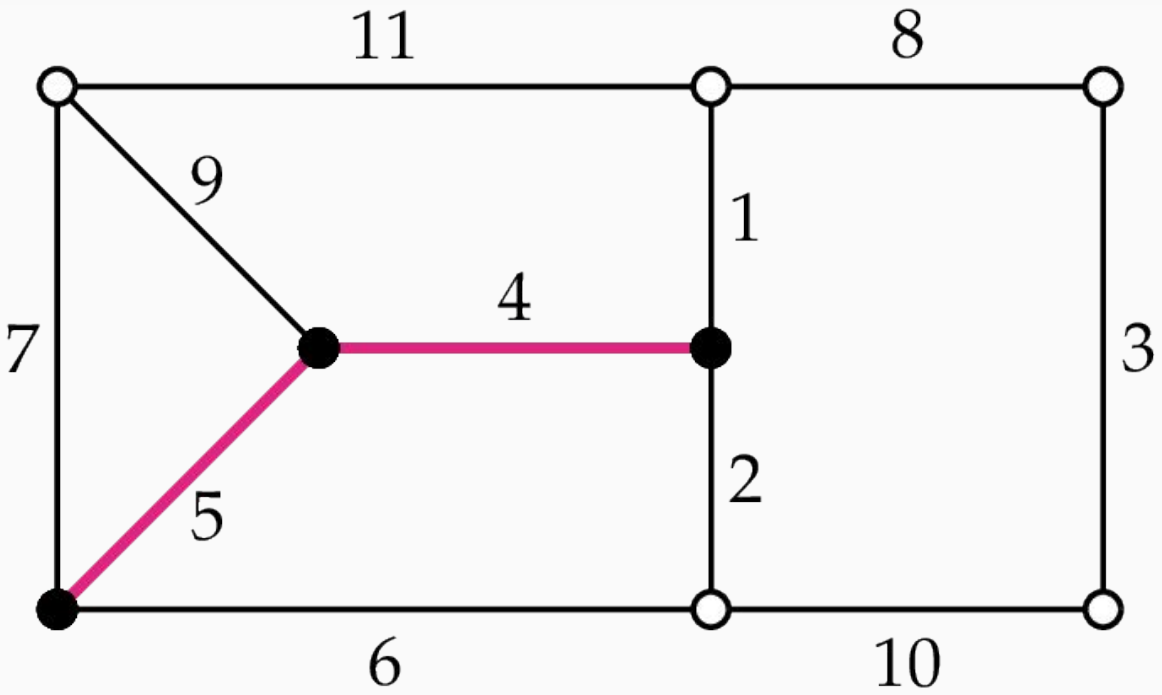

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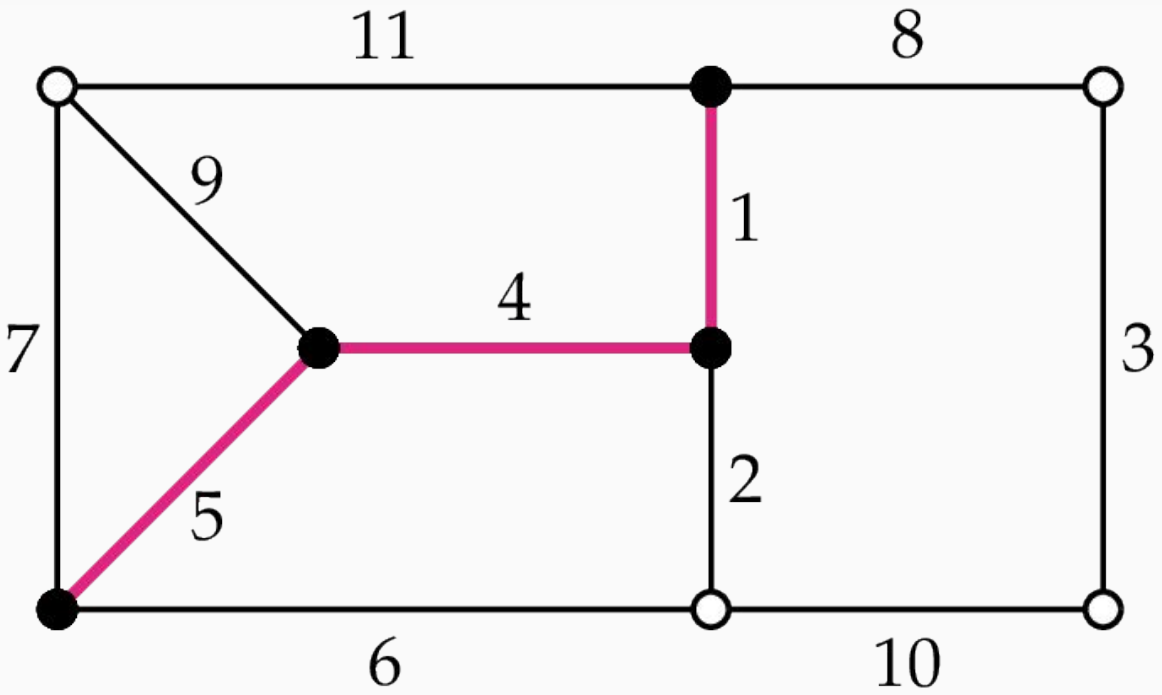

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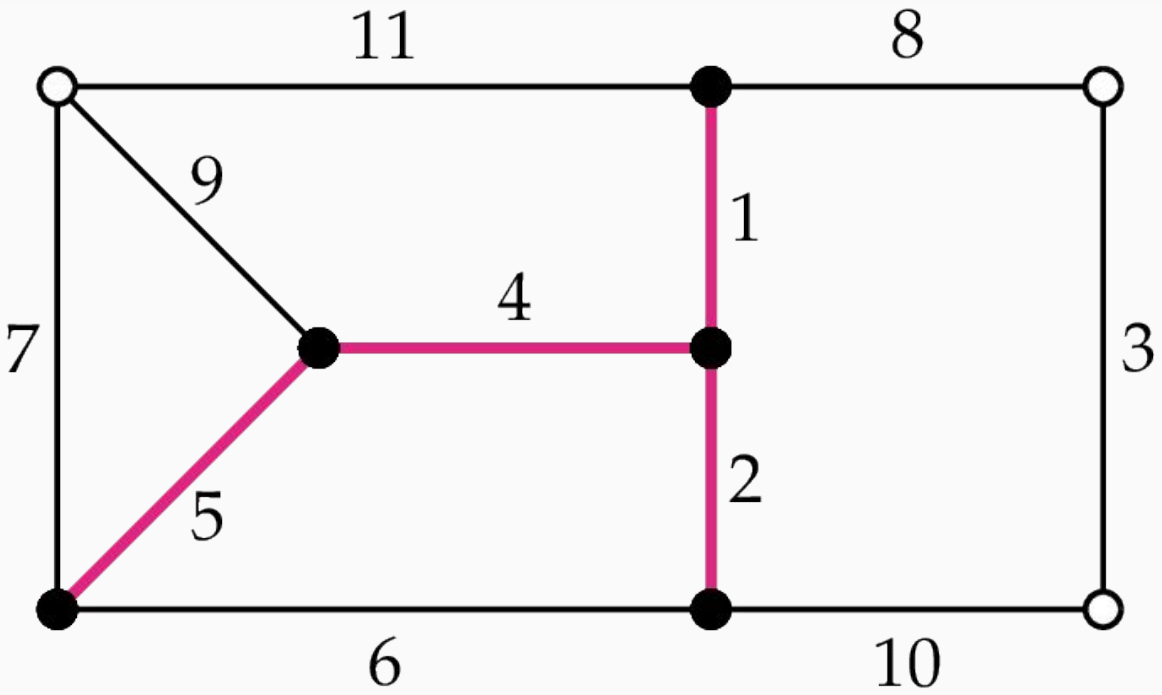

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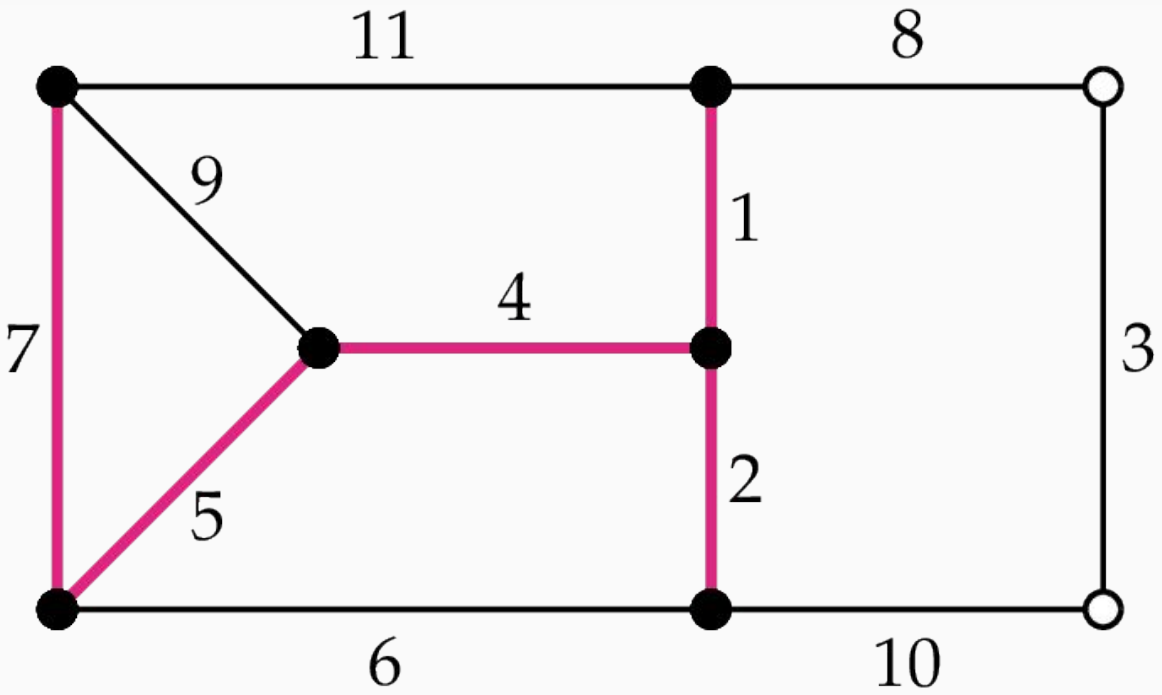

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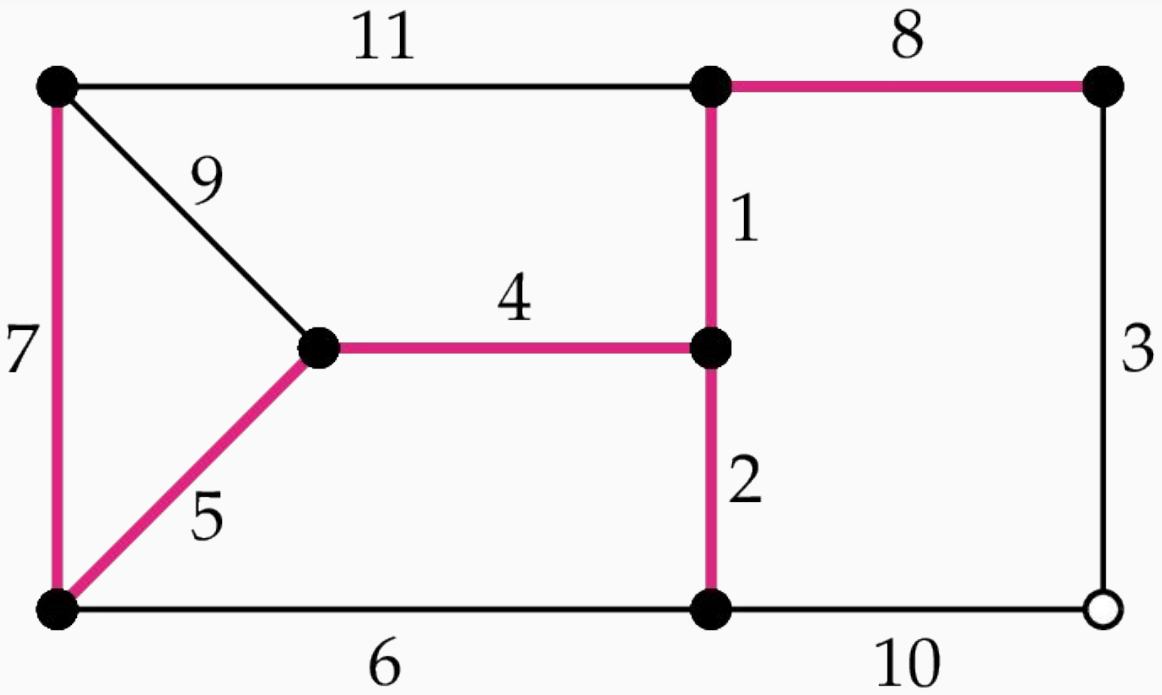

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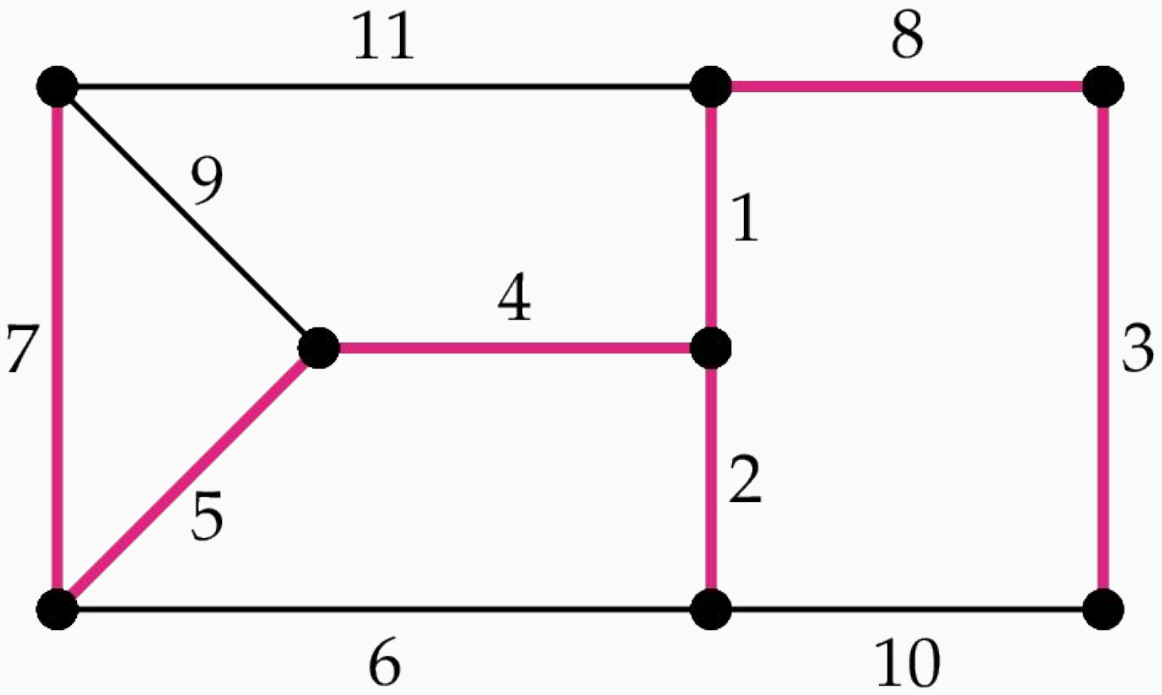

Boruvka’s algorithm

1. Repeat until one component:
    a. Identify each component’s safe edge.
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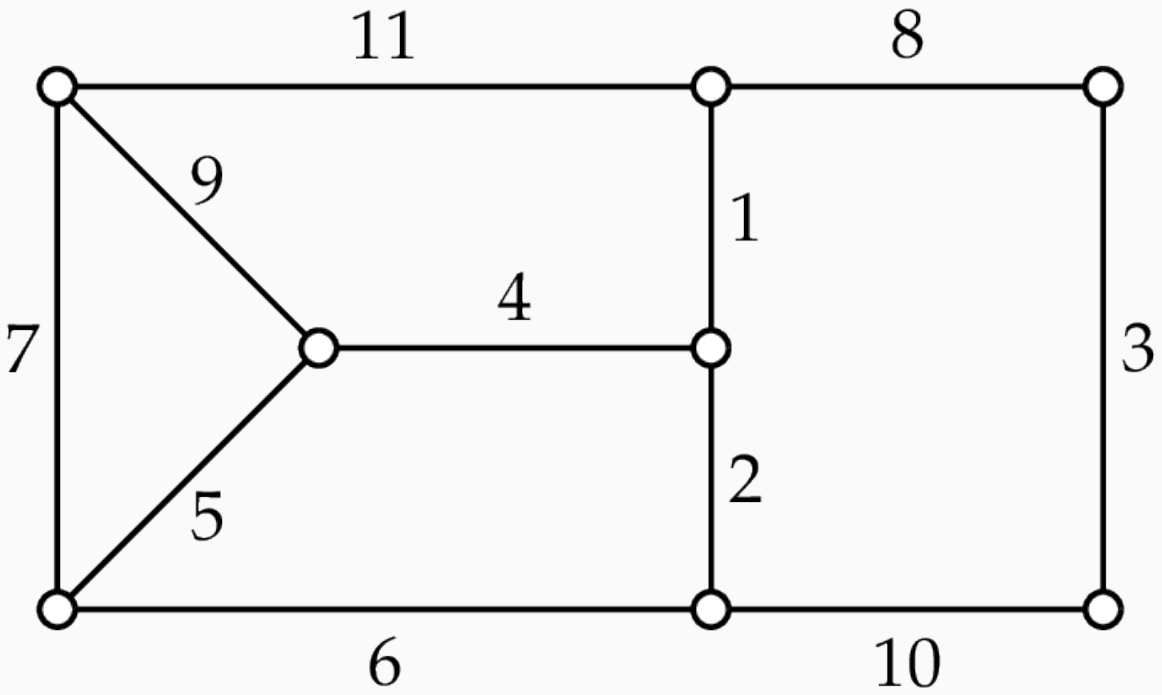

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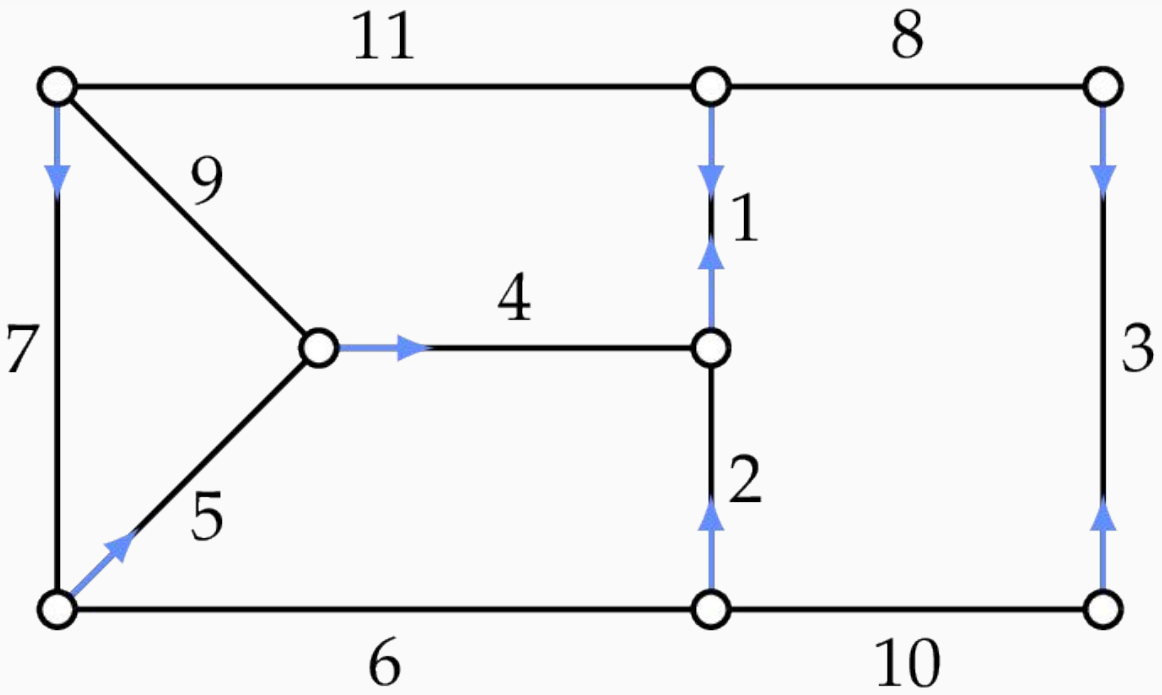

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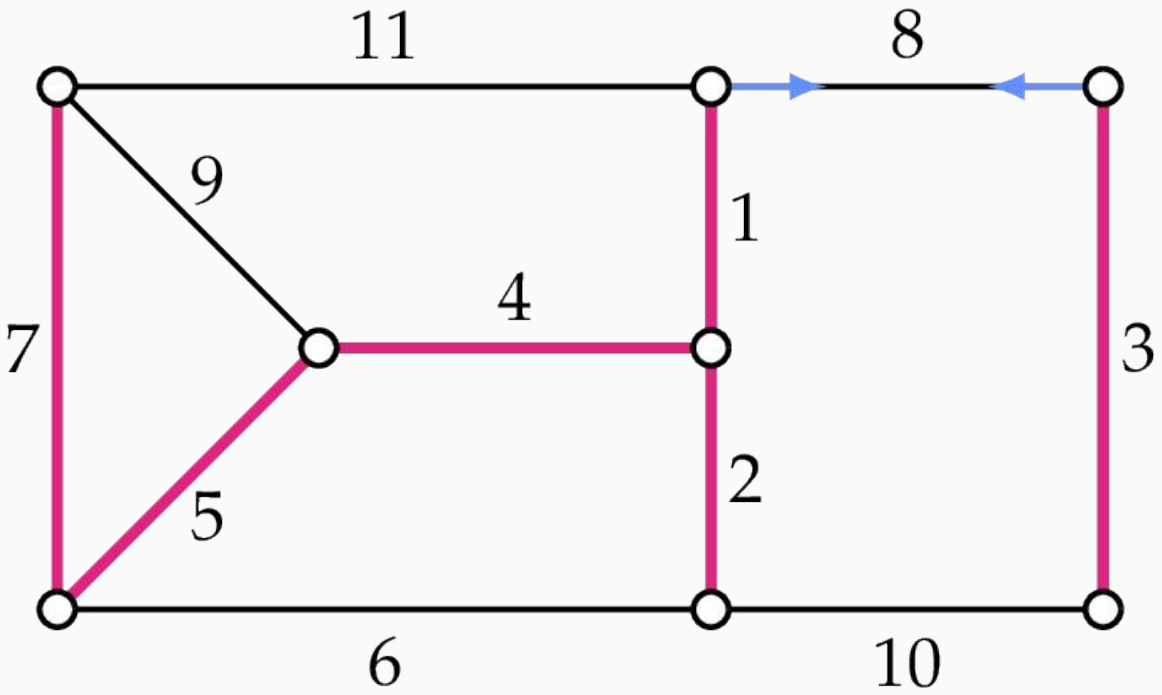

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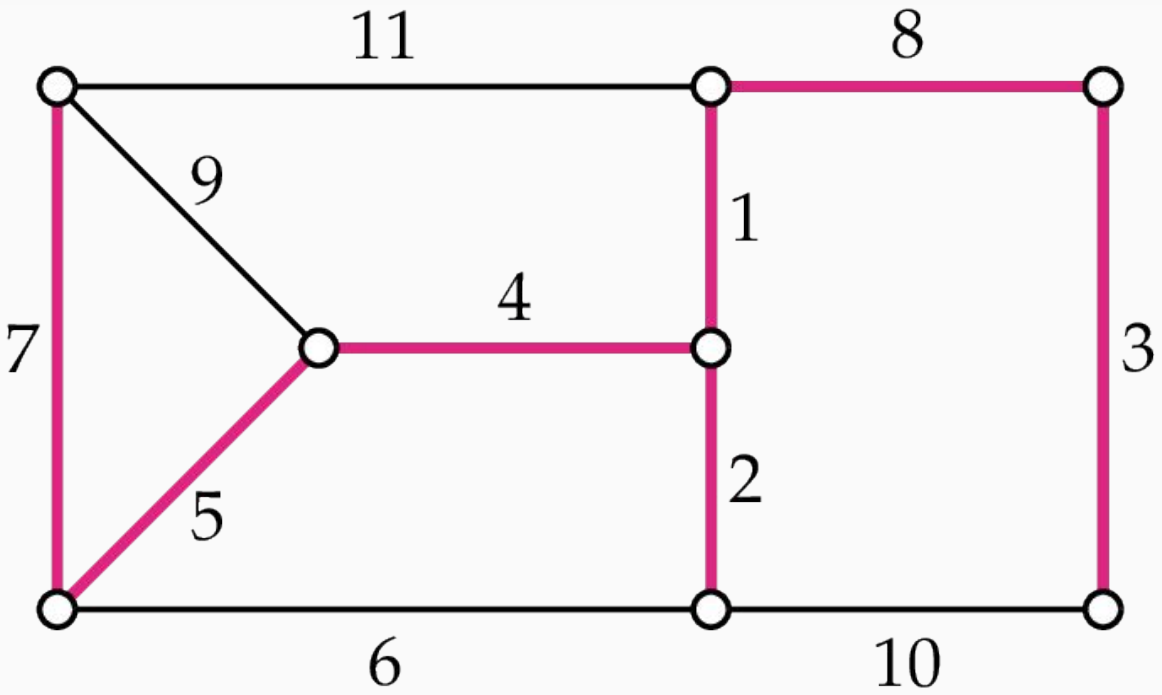

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   b. Add all safe edges.
   c. **Relabel components of forest.** (CountComp using DFS, etc.)

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Store safe edges in array indexed by component number.

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Linear scan through edges (***u***, ***v***):

- "Are ***u*** and ***v*** in different components?"

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Linear scan through edges (***u***, ***v***):

- "Are ***u*** and ***v*** in different components?"
  - Check if they're the new minimum for both components.

Boruvka's algorithm - finding safe edges

$\textsc{AddSafeEdges}(E, F, count)$:

initialize $S[1, \ldots, count]$

for each $(u, v) \in E$

if $comp(u) \neq comp(v)$

if $w(u, v) < w(S[comp(u)])$

$S[comp(u)] = (u, v)$

if $w(u, v) < w(S[comp(v)])$

$S[comp(v)] = (u, v)$

for $i = 1, \ldots, count$

add $S[i]$ to F

Boruvka’s algorithm

1. Repeat until one component:
    a. Identify each component’s safe edge. (linear scan)
    b. Add all safe edges. (adding ***k*** edges)
    c. Relabel components of forest. (DFS or something)

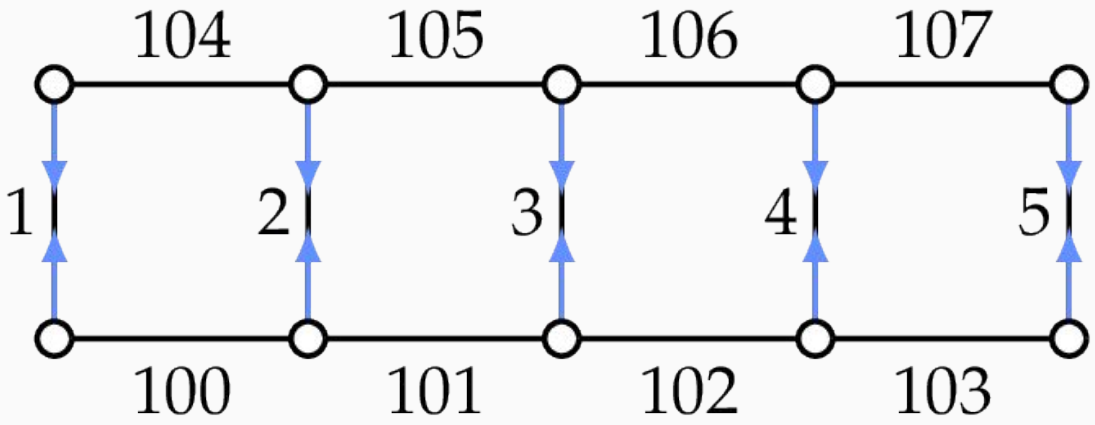
Boruvka's algorithm

1. Repeat until one component:
    a. Identify each component's safe edge. (linear scan) $O(|\boldsymbol{E}|)$
    b. Add all safe edges. (adding $\boldsymbol{k}$ edges) $O(|\boldsymbol{V}|)$
    c. Relabel components of forest. (DFS or something) $O(|\boldsymbol{E}|)$

Each iteration takes $O(|\boldsymbol{E}|)$ time. How many iterations?

Slowly shrinking components

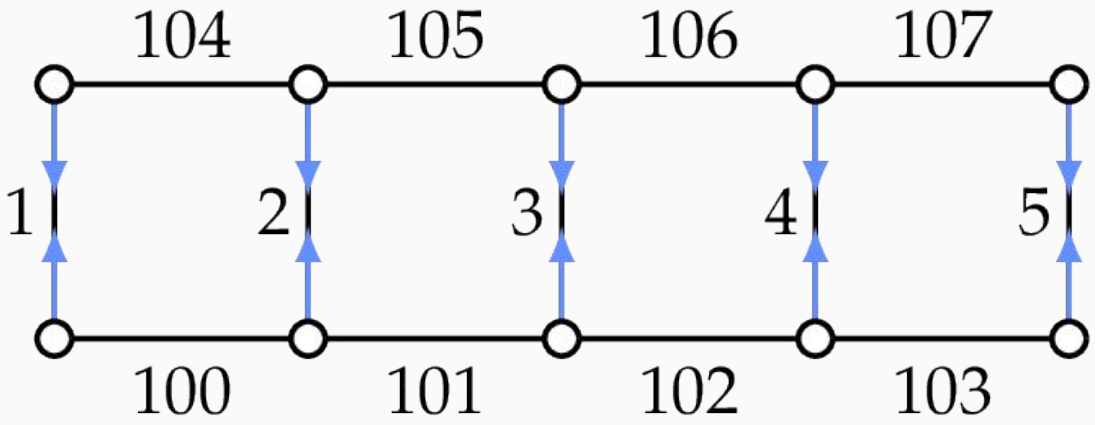
In the worst case, components are paired up, decreasing by a factor of 2.



Slowly shrinking components

In the worst case, components are paired up, decreasing by a factor of 2.

How long until there's only one component left?

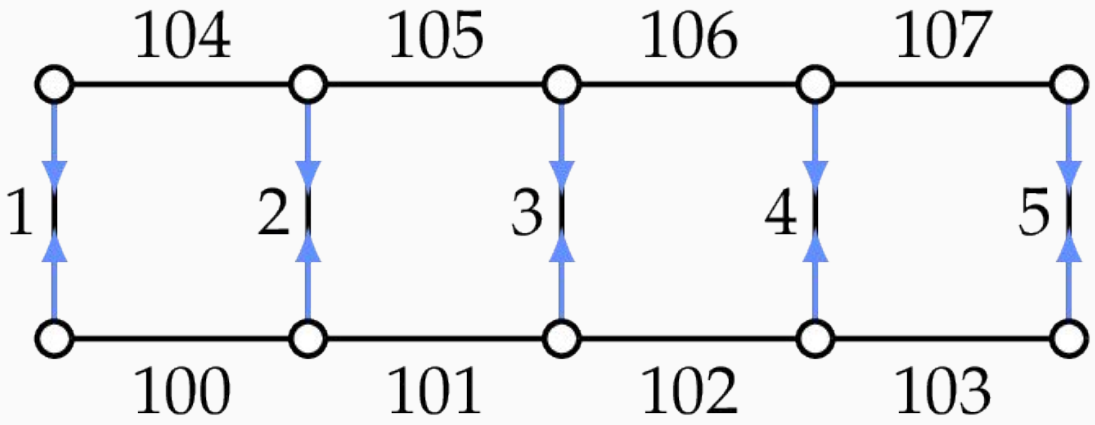


Slowly shrinking components

In the worst case, components are paired up, decreasing by a factor of 2.

How long until there's only one component left?

O(log |*V*|) iterations



Boruvka's algorithm

1. Repeat until one component:
    a. Identify each component's safe edge. (linear scan) **O(|*E*|)**
    b. Add all safe edges. (adding ***k*** edges) **O(|*V*|)**
    c. Relabel components of forest. (DFS or something) **O(|*E*|)**

Each iteration takes O(|***E***|) time. How many iterations? **O(log |*V*|)**

Total runtime: O(|***E***| log |***V***|)

Kruskal’s algorithm

1. Sort all the edge weights.
2. Repeatedly pick the smallest edge **without creating a cycle**.

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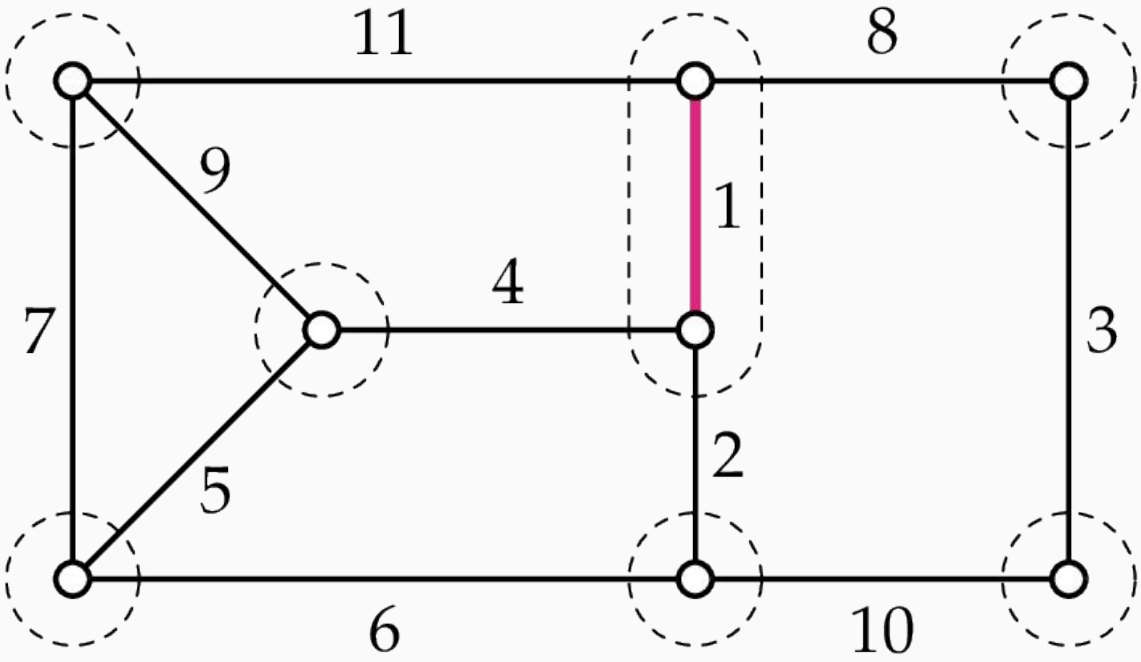
This step is about detecting components. Should we use CountComp again?

Kruskal’s algorithm

For each iteration of Kruskal, most components don’t change.



Kruskal’s algorithm

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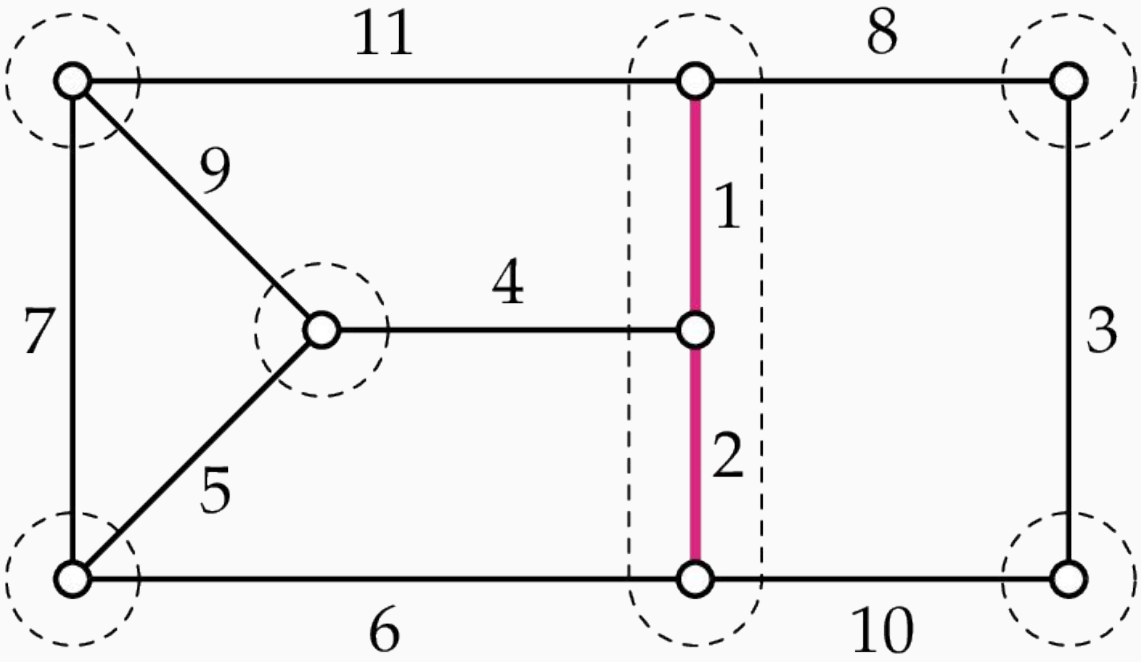
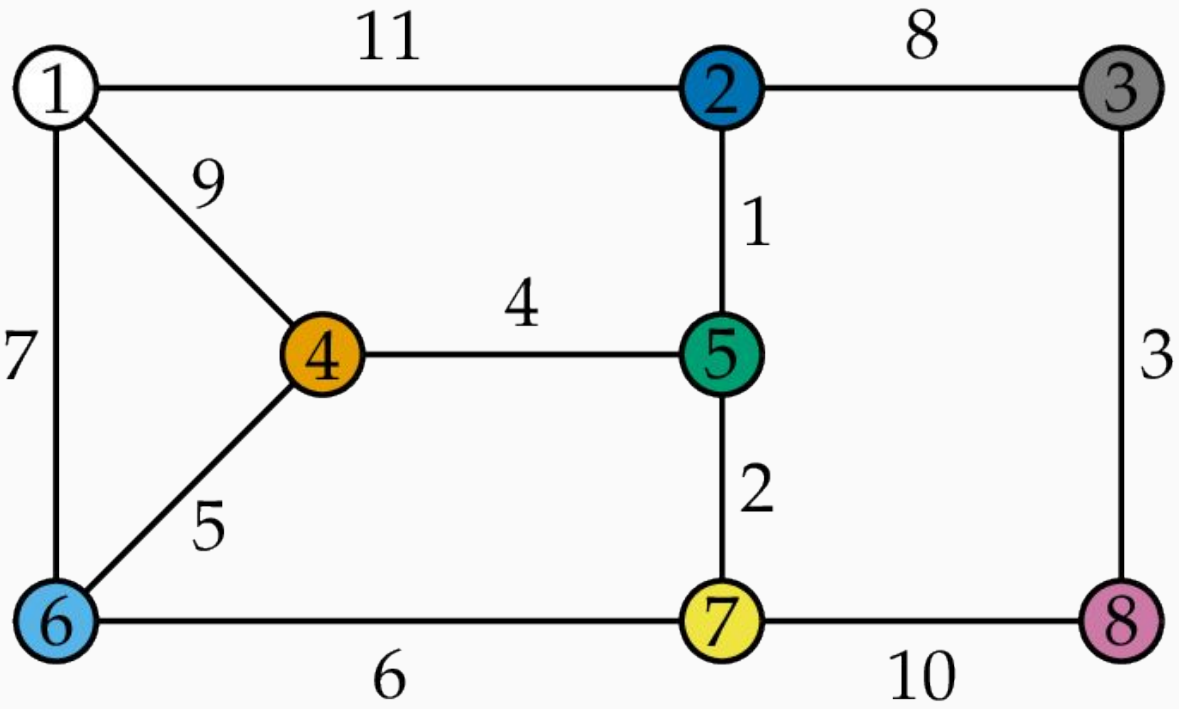


Kruskal's algorithm

Just relabel ***one of the merged components***.



Kruskal's algorithm

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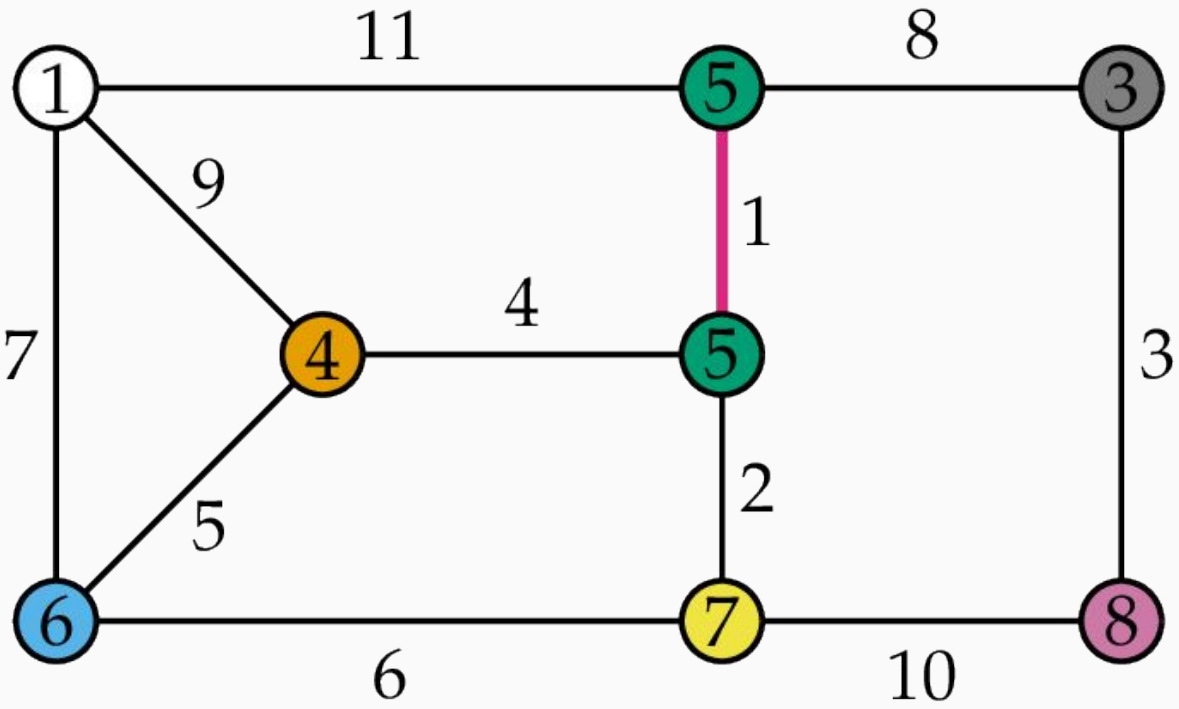
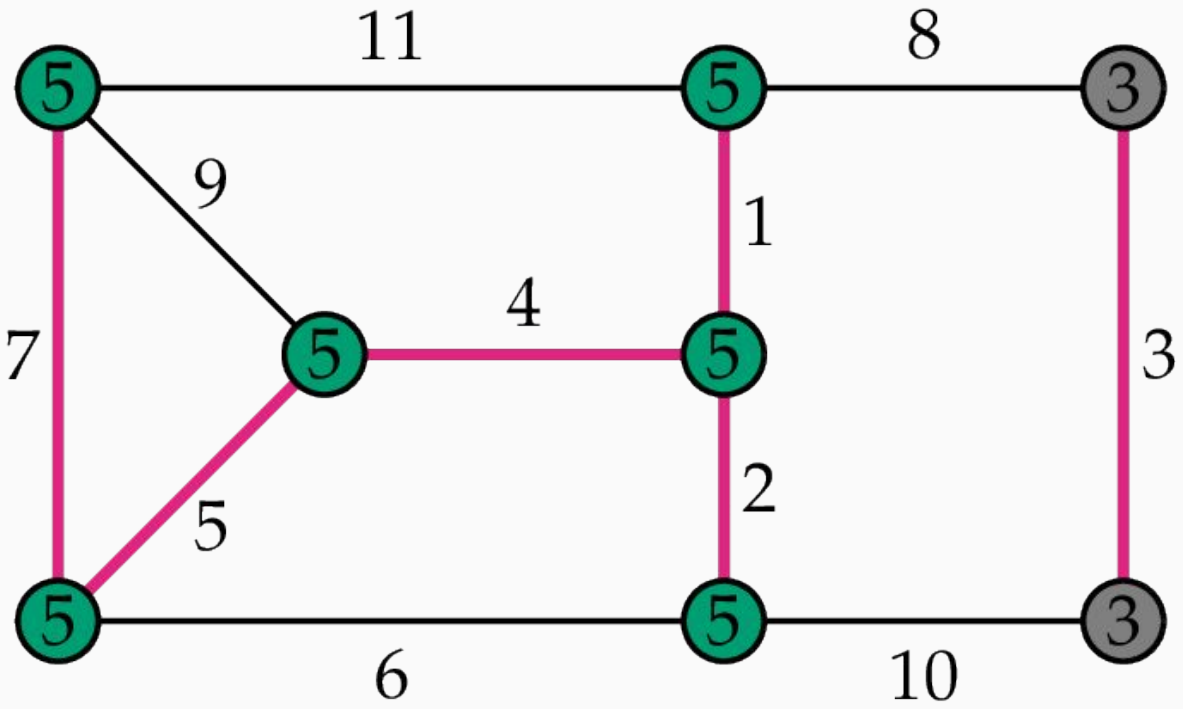


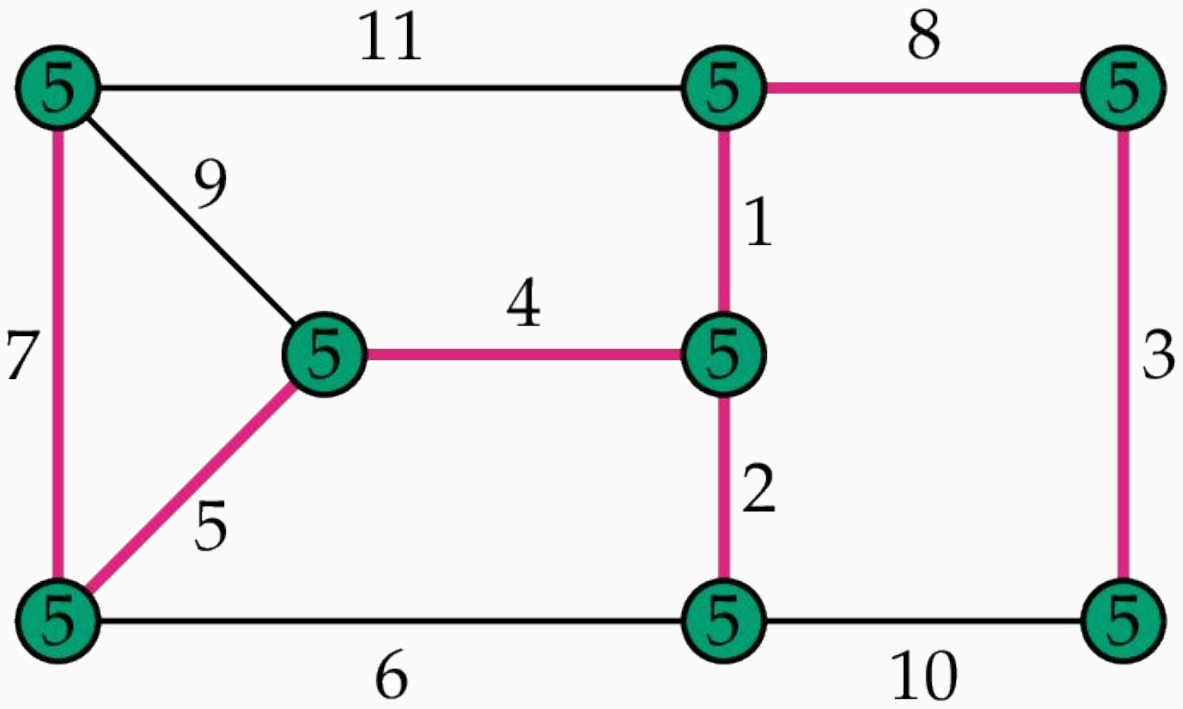
Kruskal's algorithm

Just relabel ***one of the merged components***.



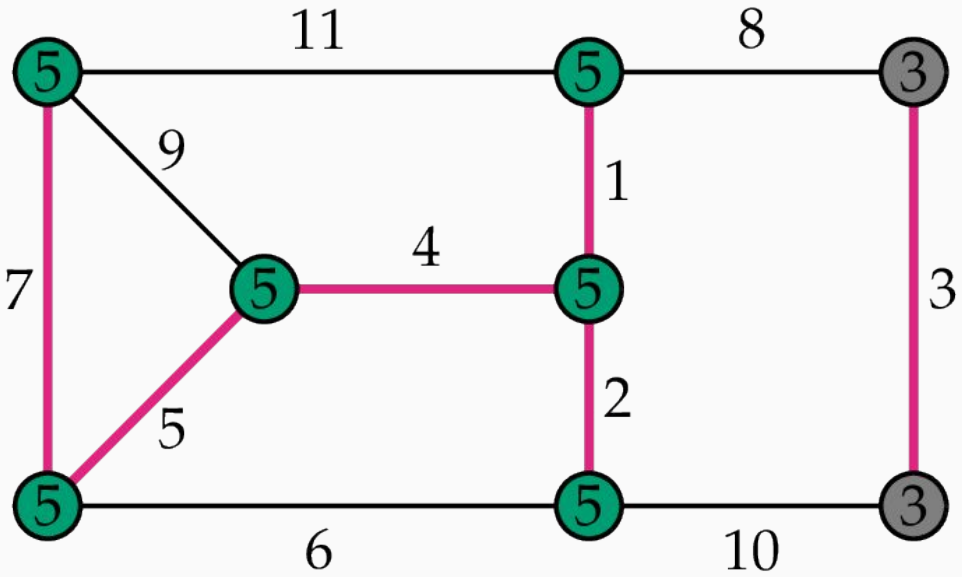
Kruskal's algorithm

Just relabel ***smaller of the two merged components***.



Kruskal’s algorithm

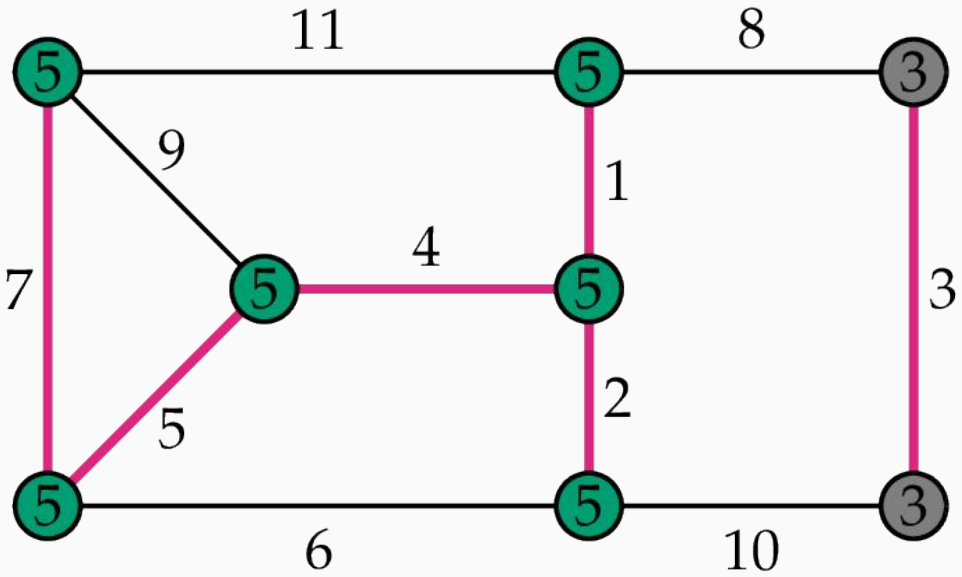
How long does relabeling take?



Kruskal's algorithm

How long does relabeling take?

- How many times can a vertex be relabeled?

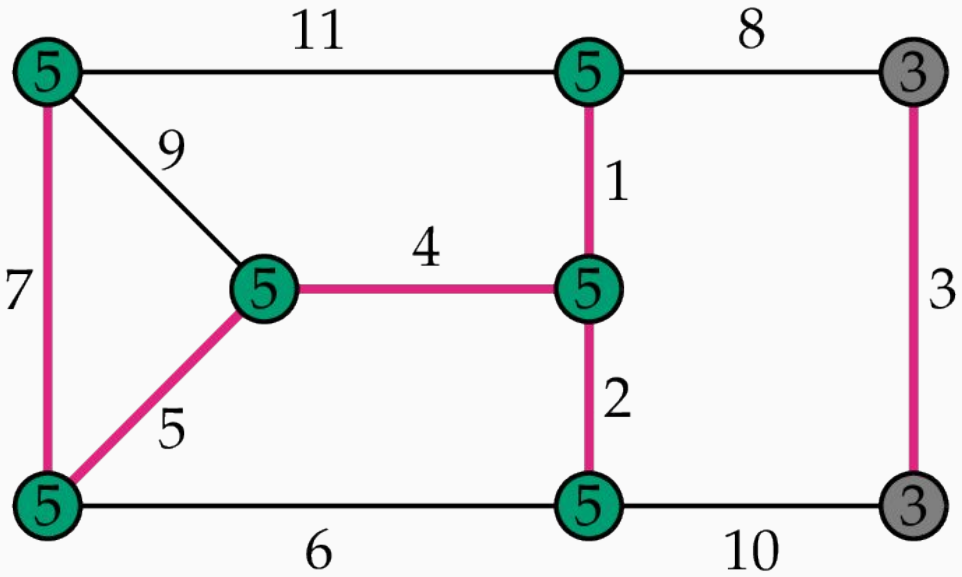


Kruskal's algorithm

How long does relabeling take?

- How many times can a vertex be relabeled? **O(log |*V*|) times**

O(|***V***| log |***V***|) in total



Kruskal’s algorithm

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1. Sort all the edge weights. $O(|\boldsymbol{E}| \log |\boldsymbol{E}|)$
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Kruskal's algorithm

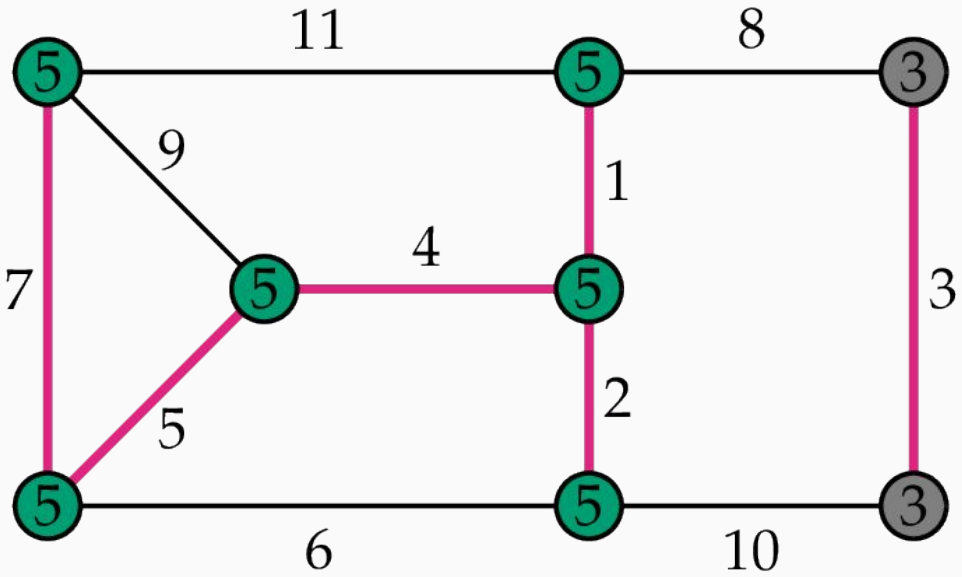
1. Sort all the edge weights. $O(|\boldsymbol{E}| \log |\boldsymbol{E}|)$
2. Repeatedly pick the smallest edge **without creating a cycle**. $O(|\boldsymbol{E}| + |\boldsymbol{V}| \log |\boldsymbol{V}|)$

Total: $O(|\boldsymbol{E}| \log |\boldsymbol{V}|)$

Kruskal's algorithm

Kruskal's algorithm often described using a **disjoint-set data structure**:

- MakeSet(***v***)
- Find(***v***)
- Union(***u***, ***v***)



Kruskal's algorithm

```
KRUSKAL(V, E):
    sort E by weight
    F := GRAPH(V, ∅)
    for each v ∈ V
        MAKESET(v)
    for each (u, v) ∈ E
        if FIND(u) ≠ FIND(v)
            UNION(u, v)
            add (u, v) to F
    return F
```

Prim’s algorithm

1. Pick any vertex **v**, and set **T** = {**v**}
2. Repeatedly “grow” **T** by adding smallest edge leaving **T**

Prim’s algorithm

1. Pick any vertex **v**, and set **T** = {**v**}
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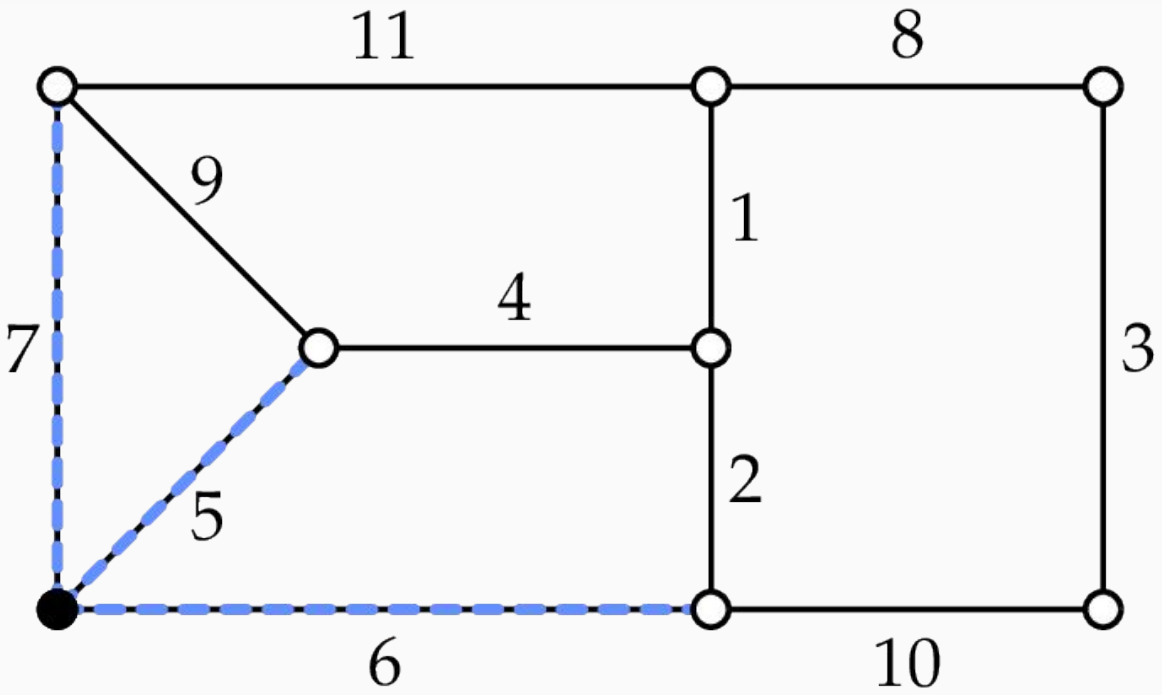
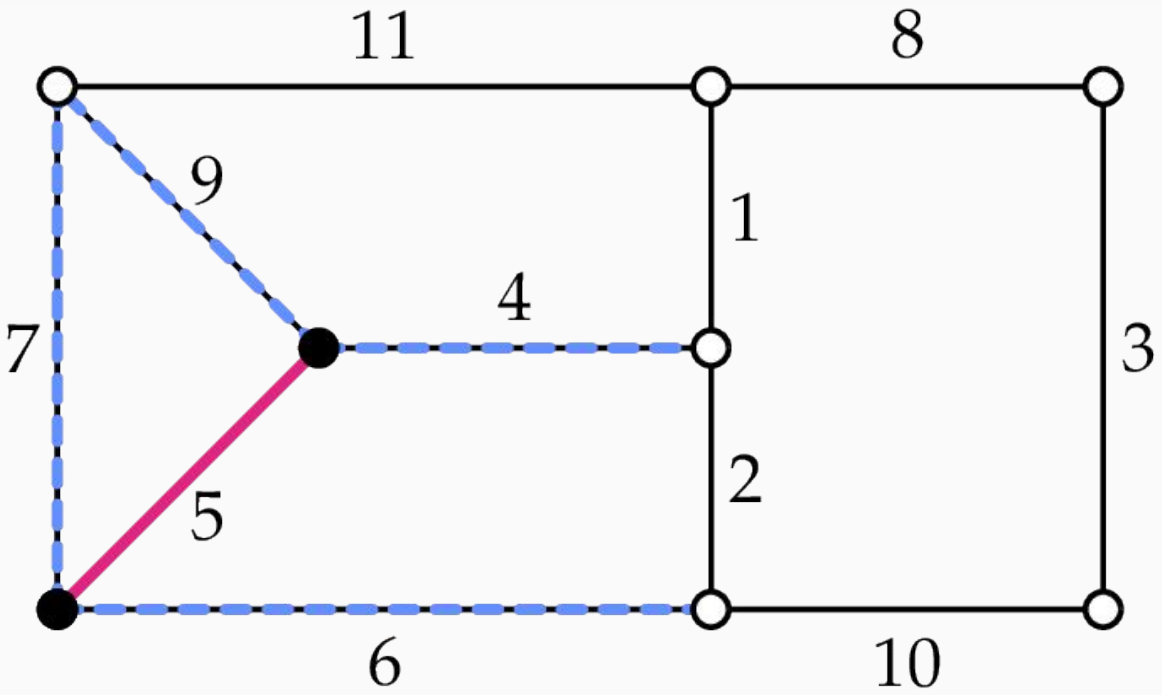
    a. Something-first search where bag = priority queue storing edges

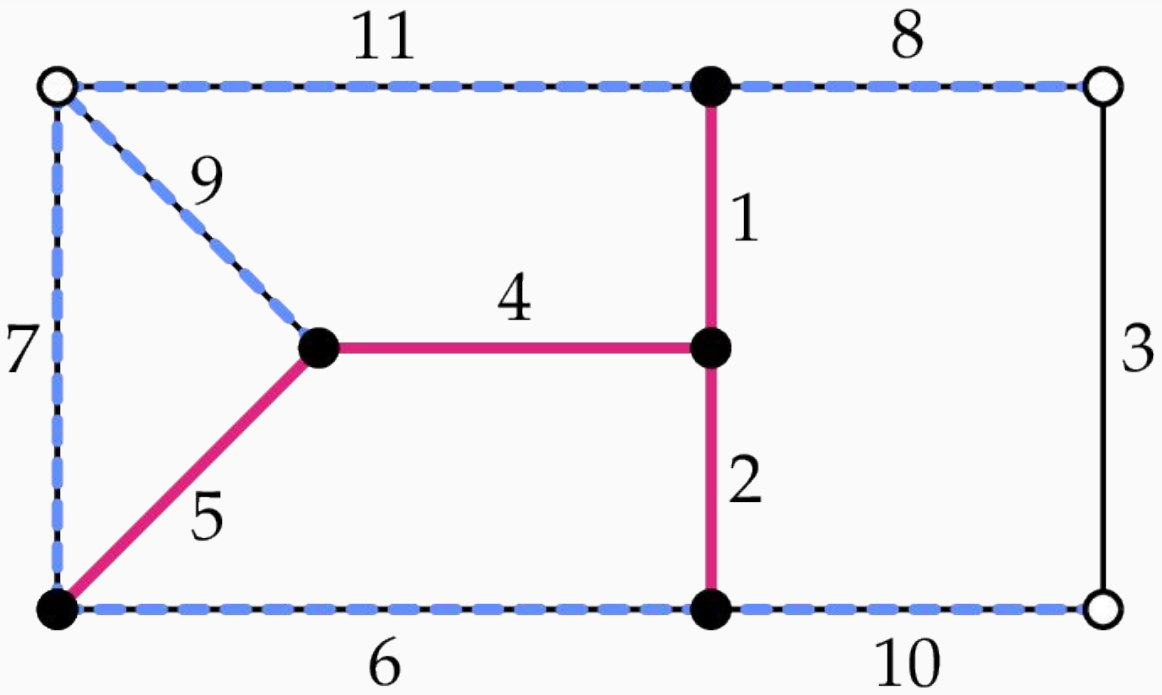
Prim’s priority queue



Prim’s priority queue



Prim’s priority queue



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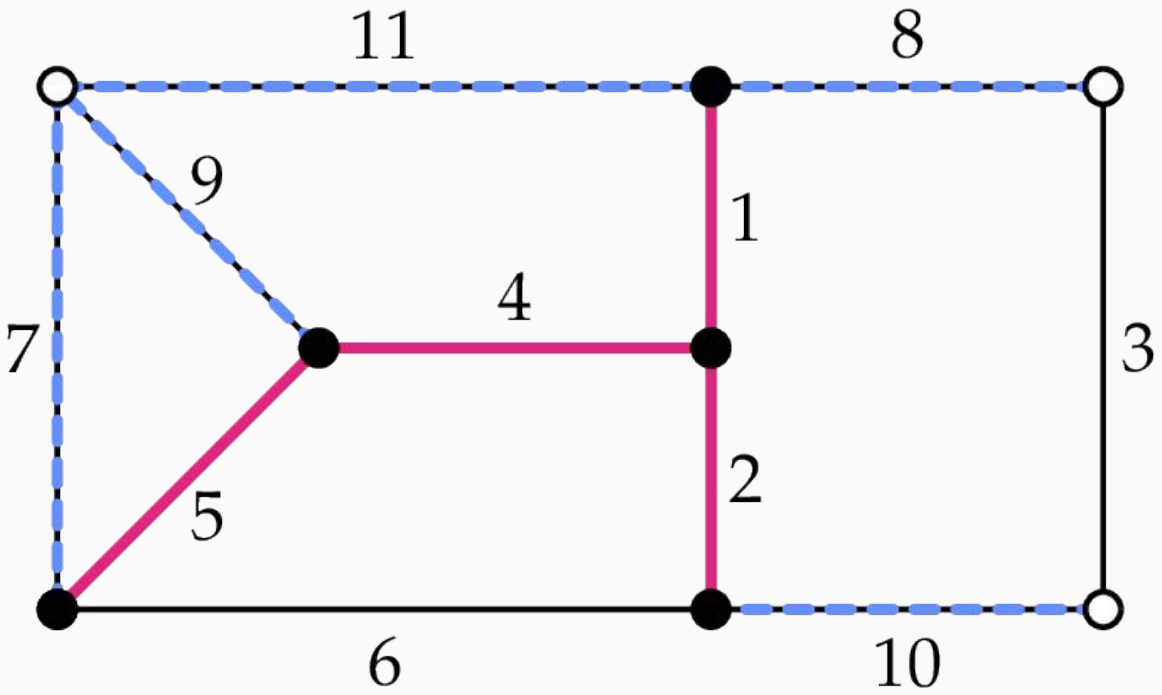
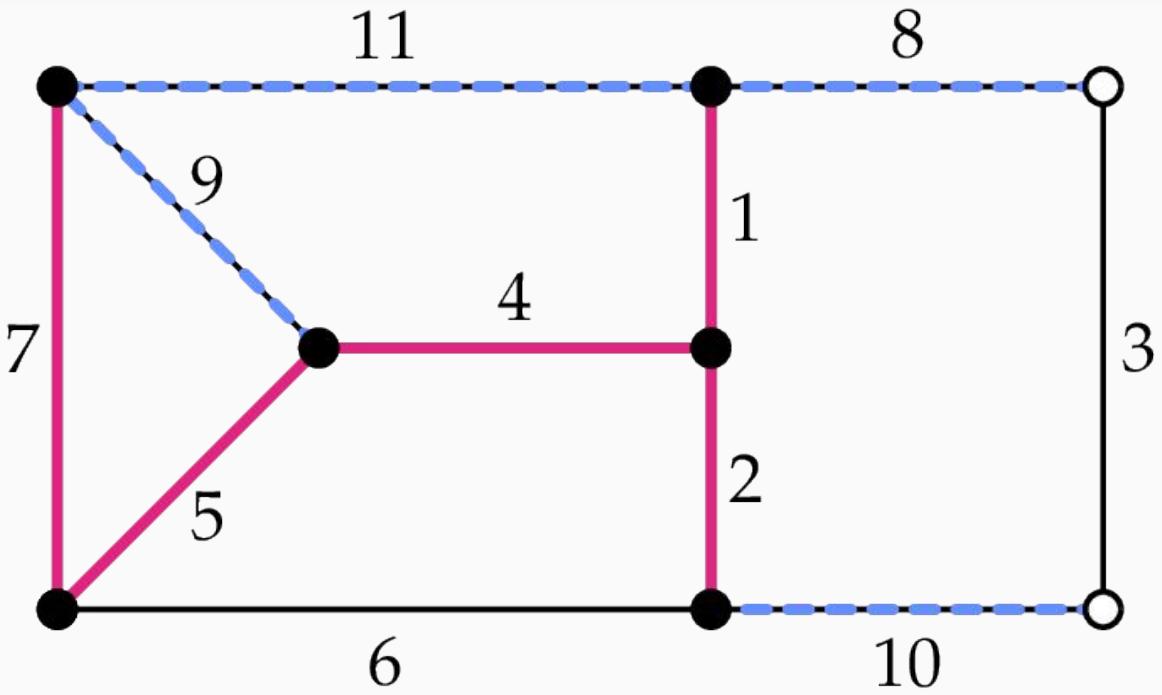


Prim’s priority queue



Prim's algorithm

```
PRIM(V, E):
    mark arbitrary vertex r
    H := HEAP()
    for each (r, v) ∈ E
        H.INSERT((r, v), w(r, v))
    F := GRAPH(V, ∅)
    while |H| > 0
        (u, v), w := H.EXTRACTMIN()
        if v is unmarked
            mark v
            add (u, v) to F
            for each (v, v′) ∈ E
                H.INSERT((v, v′), w(v, v′))
    return F
```

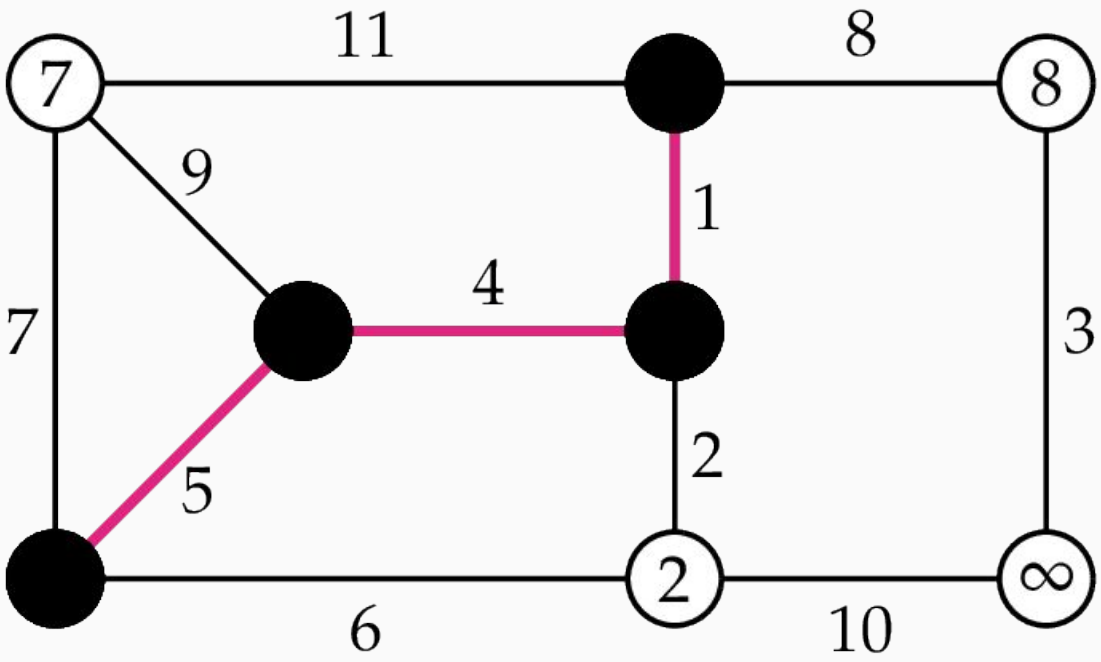
Prim's algorithm

1. Pick any vertex **v**, and set **T** = {**v**}
2. Repeatedly "grow" **T** by adding smallest edge leaving **T**
    a. Something-first search (bag = priority queue/heap)

O(|**E**| log |**V**|) as well

A vertex-centric version

An alternate implementation of Prim's algorithm uses ***vertex*** priorities



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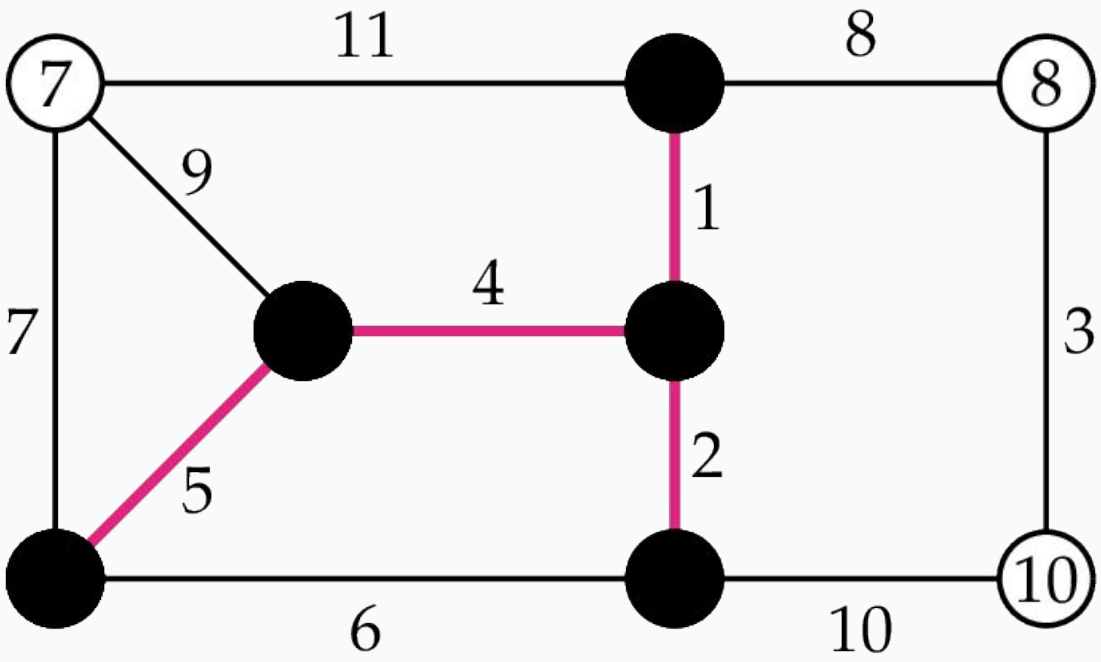

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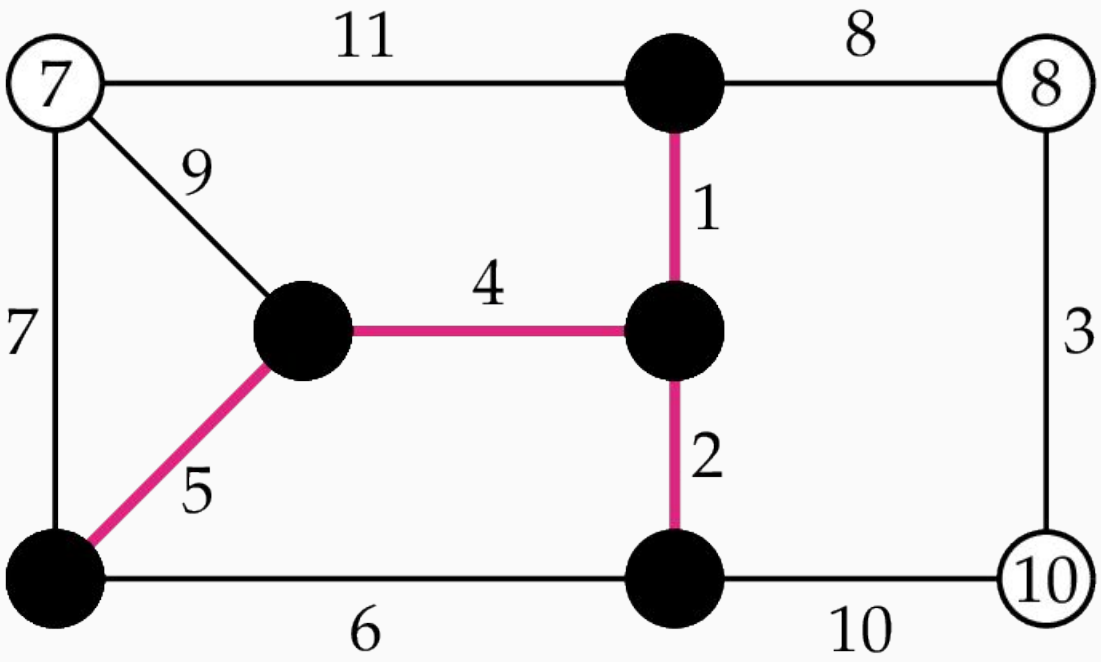
We'll see almost the same idea in Dijkstra's algorithm